BCSE498J – Project - II

QFE-COD: CAMOUFLAGED OBJECT DETECTION USING QUANTUM FREQUENCY ENHANCEMENT WITH Q-WAVEKAN-BASED FEATURE LEARNING

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

Reg. No. 22BCE0905 DAKSHETH G

Under the Supervision of

SATHYA K

Assistant Professor Sr. Grade 2

SCOPE

April 2026

<u>DECLARATION</u>

We hereby declare that the project entitled **“QFE-COD: CAMOUFLAGED OBJECT DETECTION USING QUANTUM FREQUENCY ENHANCEMENT WITH Q-WAVEKAN-BASED FEATURE LEARNING”** submitted by us, for the award of the degree of *Bachelors of Technology in Computer Science & Engineering* to VIT is a record of bonafide work carried out by us under the supervision of Prof. Dr. Sathya K.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place : Vellore

Date :

Signature of the Candidate

CERTIFICATE

This is to certify that the project entitled **"QFE-COD: Camouflaged Object Detection Using Quantum Frequency Enhancement with Q-WaveKAN-Based Feature Learning"** submitted by Daksheth G (22BCE0905), School for Computer Science and Engineering, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by the student under my supervision during Winter Semester, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute orany other institute or university. The project fulfills the requirements and regulations ofthe University and in my opinion meets the necessary standards for submission.

Place : Vellore

Date :

Signature of the Guide

Internal Examiner **External Examiner**

Head of the Department

COMPLETION CERTIFICATE ISSUED BY THE COMPANY

(applicable for NON-CDC)

1. Offer letter:

Oneindia

November 14, 2025

Dear Daksheth G,

Subject: Internship Letter

We are pleased to offer you an internship with **Greynium Information Technologies Private Limited** for a period of **six months** starting from **January 05, 2026.**

During the internship period you will be designated as **"Intern"** and will be paid a monthly stipend of Rs. **15,000.00** and any other deductions as mandated by law is payable.

Internship offered by the organization is strictly for training purposes and does not guarantee employment with the organization.

Kindly validate the project content with the project guide at **Greynium Information Technologies Private Limited** before submitting the final project report.

Please take time to understand the applicable organization policies and you will also be required to abide by the Code of Conduct, HR policies, Confidentiality, and other regulations applicable to you.

Please sign at the bottom of this letter as your acknowledgement.

We wish you the very best for your internship at Greynium.

Yours Sincerely,

For Greynium Information Technologies Private Limited

Abuthayir B

Deputy Director

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No. 2, VRR Legacy, 4th Floor, 1st Main, 1st Block, Koramangala, Jakkasandra Extension, Bengaluru, Karnataka, India 560 034
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2. Intern – Progress or Completion letter by concerned authority:

Oneindia

Daksheth G <temp-daksheth.g@oneindia.co.in>

Internship Progress Feedback - Daksheth G

1 message

Balakrishnan N <balakrishnan.n@oneindia.co.in> 28 April 2026 at 23:37
To: Daksheth G <temp-daksheth.g@oneindia.co.in>

I am writing to provide an update on the internship progress of Mr. **Daksheth G**, a final year B.Tech student of Vellore Institute of Technology (VIT), who is currently undertaking his internship with Greynium Information Technologies Private Limited under my guidance from January 5, 2026 to April 28, 2026 (ongoing)

During his tenure with us, Daksheth has been involved in our AI/ML initiatives and has contributed to multiple ongoing projects. I am outlining his key technical contributions below:

1. Article Tag Generation
Daksheth has contributed to building systems to generate relevant tags for articles, helping improve content discoverability and classification.

2. Named Entity Recognition (NER) Classification
He has contributed to NER-based models to identify and classify key entities within articles, supporting better content structuring.

3. Data Scraping from Trends Websites
He has contributed to developing data scraping workflows to extract trending data from external sources for analysis.

4. Article Sensitivity Analysis
Daksheth has contributed to models that assess the sensitivity of articles, assisting in content moderation processes.

5. Article Virality Predictor
He has contributed to building models aimed at estimating the potential virality of articles based on available data.

6. Multilingual Translator
He has contributed to integrating multilingual translation features to support content across different languages.

Beyond his technical contributions, Daksheth has demonstrated a good learning attitude and has been able to work on assigned tasks with guidance. Over the course of the internship, he is showing steady improvement in his technical understanding and execution.

Should you require any further information or clarification, please feel free to reach out.

Thanks & Regards,

Balakrishnan N
Email: balakrishnan.n@oneindia.co.in
Project Lead
Greynium Information Technologies Private Limited

ACKNOWLEDGEMENT

I **22BCE0905 – DAKSHETH G** am deeply grateful to the management of Vellore Institute of Technology (VIT) for providing me with the opportunity and resources to undertake this project. Their commitment to fostering a conducive learning environment has been instrumental in my academic journey. The support and infrastructure provided by VIT have enabled me to explore and develop my ideas to their fullest potential.

My sincere thanks **to Dr. Jaisankar N** the Dean of the **School for Computer Science and Engineering**, for constant support and encouragement. The Dean's leadership and vision have greatly inspired me to strive for excellence. The dedication to academic excellence and innovation has been a constant source of motivation.

I express my profound appreciation **to Dr. Boominathan P** the **Head of the Department of Software Systems** for insightful guidance and continuous support. The expertise and advice provided have been crucial in shaping the direction of my project. The commitment to fostering a collaborative and supportive atmosphere has greatly enhanced my learning experience. The constructive feedback and encouragement have been invaluable in overcoming challenges and achieving my project goals.

I am immensely thankful to my project supervisor, Dr. Sathya K, for dedicated mentorship and invaluable feedback. The patience, knowledge, and encouragement have been pivotal in the successful completion of this project. The willingness to share expertise and provide thoughtful guidance has been instrumental in refining my ideas and methodologies. This support has not only contributed to the success of this project but has also enriched my overall academic experience.

I extend my heartfelt thanks to all for the guidance and support received throughout this endeavour.

EXECUTIVE SUMMARY

Camouflaged Object Detection (COD) is one of the most visual challenging tasks in computer vision: it involves segmenting objects that are purposefully or naturally hidden by blending in with their backgrounds. These objects are highly visually similar to their backgrounds in terms of colour, texture and edge features, and are thus often imperceptible to the human eye upon first glance.

In this project, we propose QFE-COD (Quantum Frequency-Enhanced Camouflaged Object Detection), a new deep learning approach which combines quantum-classical hybrid computing with multi-scale frequency domain processing to boost the detection and segmentation of camouflaged objects. The novel architecture overcomes the key weakness of existing methods, which fail to exploit the discriminative frequency-domain features of camouflaged objects foregrounds as compared to their similarly textured backgrounds.

The QFE-COD framework has three stages. In Stage 1, a binary CAM/NonCAM classifier with PVTv2-B2 backbone is trained to pre-classify images to inform the segmentation pipeline. It reaches a maximum validation accuracy of 94.23% after 20 epochs of training on the COD10K-v3 dataset. In Stage 2, the main QFE-COD segmentation model - built upon a PVTv2-B4 backbone - uses a Discrete Wavelet Transform (DWT) decomposition module to divide deep features into four frequency subbands, followed by a Frequency Self-Attention (FSA) module and an 8-qubit Hybrid Quantum Circuit Module (HQCM) implemented with PennyLane to generate quantum-enhanced frequency embeddings. These embeddings are integrated with spatial features of various scales via a Feature Pyramid Network (FPN) with simplified Mamba-inspired State Space Model (SSM) blocks. The combined feature is then decoded by a Q-WaveKAN head, a quantum-gated segmentation head with learnable wavelet-based Kolmogorov-Arnold Network (KAN) activation functions. The project also features a baseline SINetV2 pipeline for ablation and benchmarking comparisons.

The project also delves deeper into the theoretical insights of integrating quantum circuits for visual feature processing, offers algorithmic details of each module, and conducts extensive ablation studies. The findings demonstrate the effectiveness of QFE-COD in camouflaged object detection, particularly in dealing with texturally-rich, multi-scale, and poorly-defined camouflaged objects.

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List of Abbreviations

COD	Camouflaged Object Detection
QFE	Quantum Frequency Enhancement
QFE-COD	Quantum Frequency-Enhanced Camouflaged Object Detection
CNN	Convolutional Neural Network
ViT	Vision Transformer
PVT	Pyramid Vision Transformer
PVTv2	Pyramid Vision Transformer Version 2
DWT	Discrete Wavelet Transform
LL	Low-Low Frequency Subband
LH	Low-High Frequency Subband
HL	High-Low Frequency Subband
HH	High-High Frequency Subband
FSA	Frequency Self-Attention
FPN	Feature Pyramid Network
SSM	State Space Model
HQCM	Hybrid Quantum Circuit Module
QML	Quantum Machine Learning
VQC	Variational Quantum Circuit
PQC	Parameterized Quantum Circuit
KAN	Kolmogorov-Arnold Network
Q-WaveKAN	Quantum Wavelet-based KAN
IoU	Intersection over Union
Dice	Dice Coefficient
BCE	Binary Cross Entropy
MAE	Mean Absolute Error

Symbols and Notations

δf	Carrier Frequency Offset (CFO)
ε	Normalized Carrier Frequency Offset (NCFO)
I	Input Image
H, W	Image Height and Width
p(i,j)	Predicted Pixel Probability
M	Predicted Mask
G	Ground Truth Mask
IoU	Intersection over Union
Dice	Dice Coefficient
f	Detection Function
x	Input Feature Vector
q_vec	Quantum Output Vector
θ	Trainable Quantum Parameters

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND:

The field of computer vision has witnessed remarkable advances over the past decade, driven primarily by the proliferation of deep convolutional neural networks and, more recently, vision transformer architectures. Object detection and segmentation — the tasks of locating and delineating objects within images — have matured to the point where machine performance rivals or exceeds human acuity on well-defined benchmark datasets such as COCO, PASCAL VOC, and ImageNet. However, the challenge of detecting camouflaged objects remains an important and largely unsolved problem that exposes the fundamental limitations of current perception models.

Camouflage is a pervasive phenomenon in the natural world. Animals from insects to cephalopods to large mammals have evolved remarkable abilities to blend into their surroundings, making them exceedingly difficult to detect even for trained human observers. The biological mechanism underlying camouflage — termed background matching — exploits the limited discriminative capacity of visual cortical processing by aligning the surface patterns, textures, and colors of the organism with those of the background environment. Similarly, artificial camouflage is widely used in military contexts, manufacturing defect concealment, and recreational art to obscure objects from visual detection systems.

Camouflaged Object Detection (COD) refers to the computational task of identifying and pixel-accurately segmenting such concealed objects from natural images. Unlike conventional salient object detection (SOD) — which targets visually prominent, attention-grabbing regions — COD must contend with objects that are specifically designed or evolved to evade detection. The high intrinsic visual similarity between the target object and its background in terms of color histogram, texture frequency, and edge response makes this task considerably more challenging than general object detection.

Figure 1.1: Example of Camouflaged Object in Natural Environment (Crab on Sand)

1.1.1 Historical Context And Evolution Of Cod Research:

Early work in COD was largely inspired by the challenge of military target detection under camouflage, which dates to the origins of aerial reconnaissance in World War I. Early computational approaches relied on hand-crafted feature descriptors such as SIFT (Scale-Invariant Feature Transform), HOG (Histogram of Oriented Gradients), and color histograms, combined with statistical classifiers. These approaches achieved limited success due to their inability to model the complex, context-dependent visual patterns characteristic of natural camouflage.

The deep learning revolution, initiated by the breakthrough performance of AlexNet on ImageNet in 2012, transformed the landscape of computer vision. Convolutional Neural Networks (CNNs) demonstrated unprecedented ability to learn hierarchical feature representations directly from raw pixel data. Early CNN-based approaches to COD adapted architectures designed for semantic segmentation — such as FCN (Fully Convolutional Network), U-Net, and DeepLab — to the camouflage detection task with modest success.

A pivotal milestone in COD research was the introduction of the SINet (Search Identification Network) framework by Fan et al. at CVPR 2020, accompanied by the large-scale COD10K dataset. SINet introduced a biologically-inspired two-module architecture — the Search Module (SM) mimicking predator prey-searching behavior, and the Identification Module (IM) for precise localization — along with the first standardized benchmark for evaluating COD models. The subsequent SINet-V2 extension (IEEE TPAMI 2022) incorporated a Neighbor Connection Decoder (NCD) and Group-Reversal Attention (GRA) modules, establishing a new state-of-the-art baseline that has been widely used as a comparison reference.

Recent work has increasingly explored the role of frequency domain information in COD. The key insight motivating this direction is that while camouflaged objects are difficult to distinguish from background in the spatial (RGB) domain, they may exhibit distinguishable patterns in the frequency spectrum. High-frequency components encode fine edge and texture details, while low-frequency components represent coarse color and structure. By separately processing and fusing these frequency bands, models can exploit discriminative cues that are invisible in the spatial domain alone.

1.1.2 Quantum Computing In Machine Learning:

Quantum computing represents a fundamentally different computational paradigm based on the principles of quantum mechanics, including superposition, entanglement, and interference. A qubit — the quantum analogue of the classical bit — can exist in a superposition of states, enabling quantum processors to represent and manipulate exponentially large state spaces with a linearly growing number of physical qubits.

Quantum Machine Learning (QML) explores how quantum circuits can be embedded within or replace components of classical machine learning pipelines. Variational quantum circuits (VQCs), also called parameterized quantum circuits (PQCs), are particularly relevant to this goal. In a VQC, trainable rotation gates encode input data into qubit states, and entangling layers create correlations between qubits that cannot be efficiently represented on classical hardware. The expectation values of observables measured on the output qubits form the quantum model's output, which can be differentiated with respect to the circuit parameters using classical gradient estimation techniques.

The PennyLane framework by Xanadu AI provides a differentiable programming interface for quantum circuits, enabling seamless integration of quantum layers with PyTorch and TensorFlow neural networks. This hybrid quantum-classical approach — where a small quantum circuit processes compressed feature vectors and feeds its output back to a classical network — offers a principled way to explore potential quantum advantages in feature transformation without requiring full quantum hardware.

i) Quantum Advantage for Frequency-Domain Visual Tasks- A strong case for quantum computing arises when the data can be naturally expressed in the frequency domain. Traditional frequency examination of the information, i.e. A discrete Fourier transform (DFT) or discrete wavelet transform (DWT) decomposes the signal into orthogonal basis components representing oscillations, i.e. patterns at various spatial scales, orientations. The workings of quantum mechanics are based on similar logic quantum states are vectors in a Hilbert space. Quantum gates are unitary transformations carried out over that area. Quantum interference results from the constructive or destructive interference of the probabilities associated with various computational paths. Additionally, there are remarkable similarities between signal-processing in the frequency domain and linear-algebraic methods in quantum theories.

The quantum Fourier transform (QFT), for instance, performs an exact discrete Fourier transform on an n-qubit register in $O(n^2)$ gate operations, compared to $O(N \log N)$ for the classical FFT where $N = 2^n$. While the QFT cannot be used to directly read out all transformed coefficients — doing so would require exponentially many measurements — it serves as a subroutine in algorithms such as Shor's factoring and quantum phase estimation, where only specific frequency components are of interest. More relevant to the present work is the observation that variational quantum circuits (VQCs) operating on frequency-encoded inputs can learn non-linear transformations over frequency feature spaces that mix components across all basis functions simultaneously through entanglement. A classical fully-connected layer operating on an N-dimensional frequency vector requires $O(N^2)$ parameters to perform an arbitrary linear mixing. A VQC with $n = \log_2(N)$ qubits and d entangling layers achieves a comparable effective mixing with $O(nd)$ parameters, operating through quantum interference rather than explicit matrix multiplication. This parameter efficiency is particularly relevant when N is large — as is the case when processing DWT subbands of deep feature maps — and when the relevant transformations are expected to involve long-range correlations across frequency components.

ii)Why Qubit Superposition Maps Naturally to Wavelet Subbands -When a 2D feature map F = RHxW is transformed to a Discrete Wavelet Transform, it generates four frequency subbands: LL (low-low coarse approximation), LH (low-high horizontal edges), HL (high-low vertical edges), and HH (high-high diagonal textures). Each subband corresponds to a different aspect of the spatial frequency content of the feature map. This four-way decomposition is an equivalent of a 2-qubit quantum state.

A general 2-qubit state has the form= $|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$where $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$. The four basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ are the four orthogonal components of the quantum state, just like the four orthogonal subband decompositions LL, LH, HL, HH of the 2D wavelet transform. The amplitude coefficients ij determine the relative contribution of each basis state, just as the energy of each wavelet subband encodes the relative prominence of each frequency component in the original feature map.

Let's assume that we have n qubits, with an n-level wavelet decomposition. As we discussed above, an n-qubit register in superposition represents a state over 2n computational basis states, while the n-level DWT produces 3n + 1 subbands (one approximation subband and three detail subbands per level). In the QFE-COD architecture, the DWT is applied once to the deepest PVTv2-B4 feature map ($F4 \in \mathbb{R}^{Bx512xHxW}$), yielding four subbands globally average-pooled to produce an 8-dimensional frequency token.

This 8-dimensional vector is encoded into a 8-qubit quantum register as AngleEmbedding, and each component xi of the frequency token is encoded as a rotation angle: $|\psi_0\rangle = \bigotimes_{i=0}^{7} R_\gamma(x_i)|0\rangle$ (1.1)

The rotation gate $R_\gamma(\theta) = \exp(-i\theta Y/2)$ maps a scalar frequency activation to a qubit rotation about the Y-axis and places it in a superposition whose amplitude is smooth of the input. The entanglement layers create correlations between all 8 qubits simultaneously — i.e., each output qubit's expected value depends on the 8 input frequency components together. This mixing through entanglement allows one to capture intersubband interactions (i.e.g., high-frequency diagonal texture energy (HH subband) and low-frequency structural energy (LL subband), which determines the boundary of camouflage objects where texture continuity along the object-background boundary renders an object invisible.

Rather, if we use the same 8-dimensional frequency token to compute a classical MLP, we calculate purely local, coordinate-wise non-linearities in the first layer and mix in the layers following. In the quantum circuit, we have to do mixing and non-linearity simultaneously by interference, which cannot be reproduced classically for circuits of sufficient depth and entanglement.

1.2 MOTIVATION:

The primary motivation for this project stems from the observation that existing COD approaches, despite their impressive performance on standard benchmarks, share a common limitation: they primarily operate in the spatial (RGB pixel) domain and do not systematically leverage the complementary discriminative information available in the frequency domain. Camouflaged objects often differ from their backgrounds most prominently in mid-frequency features — the fine texture details and boundary oscillations that correspond to the transition zones between object and background — rather than in gross color or spatial structure.

A second motivation arises from the rapid theoretical developments in quantum machine learning. While quantum advantage has been rigorously demonstrated for specific mathematical problems such as factoring and database search, its applicability to visual perception tasks remains an open and active research question. The frequency domain representation of visual features — which is naturally expressed in terms of oscillatory basis functions analogous to quantum wavefunctions — provides an intuitively compelling connection between signal processing, wavelet analysis, and quantum circuit computation.

A third motivation concerns the emerging KAN (Kolmogorov-Arnold Network) paradigm. Classical neural networks use fixed activation functions (ReLU, sigmoid, tanh) at network nodes. KANs, inspired by the Kolmogorov-Arnold representation theorem, instead place learnable activation functions on network edges, enabling a more expressive and interpretable function approximation. Wav-KAN extends this concept by using wavelet functions as the learnable activations, enabling the network to simultaneously capture multi-scale, multi-resolution features. Integrating this capability into the segmentation head, gated by quantum frequency embeddings, creates a uniquely capable hybrid architecture for the COD task.

Finally, the COD10K dataset — the largest and most comprehensive benchmark for camouflaged object detection — provides a rigorous evaluation environment. With 10,000 images covering 78 object categories across both aquatic and terrestrial camouflage scenarios, and accompanied by instance-level ground truth annotations, it offers the scale and diversity necessary to train and evaluate architectures of the complexity proposed in this project.

1.3 SCOPE OF THE PROJECT:

The scope of this project encompasses the design, implementation, training, and evaluation of the QFE-COD framework for camouflaged object detection on the COD10K-v3 dataset. The following specific elements are within scope:

1. Design and implementation of a three-stage pipeline: (1) CAM/NonCAM binary classification, (2) quantum-frequency enhanced segmentation with the proposed QFE-COD model, and (3) parallel training of the SINetV2 baseline for comparative evaluation.
2. Implementation of the PVTv2-B4 pyramid vision transformer backbone for multi-scale feature extraction, leveraging its four hierarchical feature maps at resolutions corresponding to 1/4, 1/8, 1/16, and 1/32 of the input image.
3. Development of the DWT decomposition module using Haar wavelets to extract four frequency subbands (LL, LH, HL, HH) from the deepest PVTv2 feature map, followed by frequency self-attention pooling to produce a compact 8-dimensional frequency token.
4. Integration of an 8-qubit hybrid quantum circuit module (HQCM) using PennyLane's differentiable quantum simulator, with angle embedding and 4 layers of strongly entangling circuit operations, to process frequency tokens through quantum superposition and entanglement.
5. Design of a Feature Pyramid Network (FPN) decoder with Mamba-inspired State Space Model (SSM) blocks for sequential spatial feature processing and multi-scale fusion.

6. Implementation of the Q-WaveKAN segmentation head combining quantum gating with learnable spline-based activation functions inspired by the KAN paradigm.
7. Training on 6000 COD10K-v3 images and evaluation on 4000 test images, with performance measured using Dice coefficient and Intersection over Union (IoU) as primary metrics.
8. Comparative analysis against the SINetV2 baseline under identical training conditions.

The following are explicitly outside the scope of this project: evaluation on additional COD benchmark datasets (CAMO, CHAMELEON, NC4K); deployment of the quantum circuit on actual quantum hardware; real-time inference optimization; and application to video-based camouflaged object detection.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 LITERATURE REVIEW :

The literature on camouflaged object detection has grown substantially since the introduction of the COD10K benchmark. This section organizes existing work into five thematic clusters: traditional CNN-based approaches, transformer-based methods, frequency-domain approaches, boundary-aware methods, and quantum-inspired deep learning.

2.1 1. CNN-based Camouflaged Object Detection:

Fan et al. (2020) introduced the SINet framework alongside the COD10K dataset, establishing the first comprehensive baseline for camouflaged object detection in their paper "Camouflaged Object Detection" (CVPR 2020). SINet employed Receptive Field (RF) blocks to simulate the multi-scale receptive behavior of biological visual neurons, and a Partial Decoder Component (PDC) for identification. The COD10K dataset provided 10,000 images spanning 78 camouflaged object categories, annotated at object-level, instance-level, and with bounding boxes and attributes. Fan et al. (2022) extended SINet to SINet-V2 in IEEE TPAMI in their paper "Concealed Object Detection", introducing a Neighbor Connection Decoder (NCD) and Group-Reversal Attention (GRA) module, achieving state-of-the-art performance that became the dominant baseline in subsequent literature. The SINet-V2 achieves Dice scores above 0.85 on CAMO and approximately 0.82 on COD10K test with ResNet-50 backbone.

Feature Aggregation and Propagation Network (FAPNet) proposed by Zhou et al. in "Feature Aggregation and Propagation Network for Camouflaged Object Detection" explored multi-scale feature aggregation with refinement pathways specifically designed to handle the ambiguous boundaries characteristic of camouflaged objects. The Find Net (FindNet) approach by Li et al. in "FindNet: Can You Find Me? Boundary-and-Texture Enhancement Network for Camouflaged Object Detection" introduced boundary-and-texture enhancement, recognizing that the most challenging aspect of COD is the precise delineation of object boundaries against visually similar background textures.

ZoomNeXt by Pang et al. in "ZoomNeXt: A Unified Collaborative Pyramid Network for Camouflaged Object Detection" constructed a unified collaborative pyramid network that employs zoom-in and zoom-out strategies to leverage multi-scale contextual information at different granularities. The work demonstrated that explicitly modeling spatial scale relationships across hierarchical feature maps provides significant improvements in detecting objects of varying sizes, which is particularly relevant to COD10K where object sizes vary over three orders of magnitude.

2.1.2 Transformer-based Approaches:

The emergence of vision transformers (ViT, Swin Transformer, PVTv2) has opened new directions for COD research. Transformers are particularly well-suited to COD due to their global self-attention mechanism, which can model long-range dependencies between spatially distant regions — a critical capability when object boundaries blend with background texture across large spatial extents.

Wang et al. (2022) introduced PVT-V2 (Pyramid Vision Transformer version 2) in "PVT v2: Improved Baselines with Pyramid Vision Transformer", significantly improving over the original PVT-V1 through three key modifications: (1) linear complexity spatial reduction attention, (2) overlapping patch embedding for improved local feature continuity, and (3) convolutional feed-forward networks for enhanced spatial modeling. PVTv2-B4 in particular has emerged as a popular backbone for dense prediction tasks including COD, achieving comparable performance to Swin Transformer while maintaining more favorable computational properties.

The Visual Saliency Transformer (VST) by Liu et al. in "Visual Saliency Transformer" (ICCV 2021) demonstrated that pure transformer models can effectively handle the COD task without CNN components. The SAM-Adapter work by Chen et al. in "SAM-Adapter: Adapting Segment Anything in Underperformed Scenes" (ICCV 2023) explored adapting the Segment Anything Model (SAM) foundation model to underperformed scenarios including camouflaged objects, highlighting the emerging role of large foundation models in specialized vision tasks. The FGSA-Net (Frequency-Guided Spatial Adaptation) by Chen et al. (2024) specifically combined frequency domain information with transformer adaptation modules, showing that frequency guidance consistently improves transformer-based COD performance.

2.1. Frequency Domain Methods:

Zhong et al. (CVPR 2022) introduced the frequency domain as an explicit additional clue for COD in their paper "Detecting Camouflaged Object in Frequency Domain". Their FDNet framework designed a Frequency Enhancement Module (FEM) using offline Discrete Cosine Transform (DCT) followed by learnable enhancement operations. The key insight was that camouflaged objects, despite being visually similar to their backgrounds in the spatial domain, exhibit distinct spectral signatures that can be leveraged for detection.

Cong et al. (2023) proposed the Frequency Perception Network (FPNet) in "Frequency Perception Network for Camouflaged Object Detection" with a learnable octave convolution-based frequency perception module for coarse localization, followed by a detail-preserving refinement stage. Their approach demonstrated that online learned frequency features, as opposed to offline DCT-based features, provide better adaptability to the diversity of camouflage types encountered in COD10K.

Xie et al. (2023, ACM MM) introduced Frequency Representation Integration Network (FRINet), obtaining high-frequency components via Laplacian pyramid decomposition while using a standard transformer encoder to capture low-frequency global context. Their Frequency Representation Reasoning Module progressively eliminated discrepancies between different frequency representations through point-wise relational modeling.

Liu et al. (2022, ACM TOMM) proposed FBNet with two novel modules: a Frequency-aware Context Aggregation (FACA) module to suppress confusing high-frequency texture information, and an Adaptive Frequency Attention (AFA) mechanism. They also introduced a gradient-weighted loss function to guide the model to focus on contour details of camouflaged objects.

The FFENet by Chen et al. (2025, ScienceDirect) addressed the limitation of simple frequency feature fusion by designing a dedicated fusion module that uses cross-attention to adaptively weight high-frequency and low-frequency features. This work highlighted the complementary nature of high-frequency (edge/texture) and low-frequency (structure/color) information for COD.

The EPFDNet proposed edge-perception in the frequency domain, specifically using deep wavelet-like transformation to decompose features into different frequency bands and enhance high-frequency edge information. This work established the connection between wavelet-based frequency decomposition and boundary detection in the COD context — a connection directly exploited in the QFE-COD framework.

2.1.4 Quantum Machine Learning Background:

The theoretical foundations for quantum machine learning were established in a series of papers exploring whether quantum circuits can provide computational advantages over classical neural networks. Schuld et al. demonstrated that variational quantum circuits with angle embedding can be understood as kernel methods in a high-dimensional Hilbert space, providing a theoretical framework for understanding their expressibility. Cerezo et al. (2021) provided a comprehensive survey of variational quantum algorithms and their training challenges in "Variational Quantum Algorithms" (Nature Reviews Physics), including the barren plateau problem whereby gradients vanish exponentially in circuit depth.

Hybrid quantum-classical architectures — where quantum circuits are embedded as layers within classical neural networks — have been explored for image classification, natural language processing, and time series analysis. The PennyLane framework enables automatic differentiation through quantum circuits using the parameter-shift rule, enabling end-to-end gradient-based optimization of hybrid models.

Wang et al. and Cherrat et al. explored quantum approaches to image processing, demonstrating that quantum circuits can function as feature transformers that leverage quantum interference to produce transformations not easily realizable with classical operations of comparable complexity. While quantum hardware constraints currently limit practical deployment, simulator-based hybrid models have shown promising results in controlled experiments.

2.1.5 Kolmogorov-Arnold Networks (KAN):

Liu et al. (2024) introduced Kolmogorov-Arnold Networks (KANs) in "KAN: Kolmogorov-Arnold Networks" as an alternative to the conventional multilayer perceptron (MLP) paradigm. While MLPs place fixed activation functions at network nodes and learn weights on edges, KANs place learnable univariate activation functions on network edges, with nodes performing simple summation. This design is inspired by the Kolmogorov-Arnold representation theorem, which states that any multivariate continuous function can be decomposed into compositions and sums of univariate continuous functions.

Bozorgasl and Chen (2024) proposed Wav-KAN in "Wav-KAN: Wavelet Kolmogorov-Arnold Networks", which instantiates KAN activation functions as wavelet basis functions rather than B-splines. This modification provides several advantages: wavelet functions are inherently multi-scale, providing simultaneous representation of coarse and fine signal features; they maintain orthogonality properties that improve numerical stability; and they produce compact, interpretable representations that can be analyzed in terms of scale and frequency. Wav-KAN demonstrated superior accuracy and faster training compared to B-spline KAN (Spl-KAN) and traditional MLPs on benchmark tasks.

The conceptual alignment between wavelet-based KANs and frequency domain image analysis makes Wav-KAN a natural choice for the segmentation head in a frequency-augmented COD architecture. The Q-WaveKAN head in the proposed framework extends this concept by introducing quantum gating — where the quantum frequency embedding modulates the KAN activation magnitudes — creating a direct information pathway from the quantum frequency processing stage to the final segmentation output.

Table 2.1: Summary of Existing Camouflaged Object Detection Methods

Category	Paper Title	Authors & Year	Key Idea	Contribution	Limitation
CNN-based	Camouflaged Object Detection	Fan et al., 2020	RF + Partial Decoder	Introduced COD10K benchmark	Limited global context
CNN-based	Concealed Object Detection	Fan et al., 2022	NCD + GRA	Strong SINet-V2 baseline	CNN limitations

CNN-based	Feature Aggregation and Propagation Network for Camouflaged Object Detection	Zhou et al., 2022	Multi-scale feature refinement	Handles ambiguous boundaries	No frequency awareness
CNN-based	ZoomNeXt: Unified Collaborative Pyramid Network for COD	Pang et al., 2023	Zoom-in/out pyramid strategy	Multi-scale context modeling	No quantum or frequency component
Transformer	PVTv2: Improved Baselines with Pyramid Vision Transformer	Wang et al., 2022	Hierarchical transformer	Efficient global modeling	No frequency modeling
Transformer	Visual Saliency Transformer	Liu et al., 2021	Pure transformer	Removes CNN	High cost
Transformer	Uncertainty-Guided Transformer for Camouflaged Object Detection	Yang et al., 2023	Uncertainty estimation	Handles low-confidence regions	No frequency branch
Frequency	Detecting Camouflaged Object in	Zhong et al., 2022	DCT features	Uses spectral info	Static features

	Frequency Domain				
KAN	KAN: Kolmogorov -Arnold Networks	Liu et al., 2024	Learnable functions	MLP alternative	New method
Quantum ML	Quantum Machine Learning	Biamonte et al., 2017	QML foundations	Established quantum-classical hybrid framework	Theoretical, no vision tasks
Quantum ML	Variational Quantum Algorithms	Cerezo et al., 2021	VQC optimization	Trainable quantum circuits via parameter-shift	Noise sensitivity on real hardware

2.2 RESEARCH GAP:

2.2.1 Frequency-spatial integration remains shallow:

Existing frequency-domain COD methods such as FDNet and FPNet treat frequency features as auxiliary inputs fused through simple concatenation or addition, without modeling cross-modal dependencies between frequency bands and spatial regions. FFENet introduces cross-attention between high and low frequency streams but does not consider quantum-statistical structure of frequency distributions. QFE-COD addresses this by using quantum entanglement to model complex inter-frequency correlations that classical attention mechanisms cannot express.

2.2.2 Processing of visual features is novel for COD:

There has been no previous work on quantum circuits for frequency feature processing in any dense prediction or COD task, even though quantum entanglement has the theoretical capacity to encode inter-frequency correlations which would require an exponentially large number of classical parameters. Quantum-classical hybrid models have been proposed for image classification, but have not yet been applied in frequency feature transformation for segmentation. QFE-COD breaks this frontier with wavelet coefficients as angles rotating qubits processed through highly entangling layers using PennyLane.

2.2. KANs have not been used for dense prediction:

Current KAN and Wav-KAN frameworks only work on global classification and regression problems, while pixel-level dense prediction tasks (such as semantic segmentation and COD) remain unexplored. Dense prediction challenges, such as spatial coherence, resolution preservation, and multi-scale context, are unaddressed by existing KAN models. QFE-COD's Q-WaveKAN head is the first KAN-based segmentation head, which extends Wav-KAN with quantum gating for adaptive frequency-based pixel prediction.

2.2.4. Pre-filtering of CAM vs NonCAM images:

Existing COD pipelines process all images uniformly regardless of whether camouflaged objects are present, leading to unnecessary computation and elevated false positive rates on non-camouflaged backgrounds. No prior work incorporates a dedicated pre-classification stage to route images before segmentation. QFE-COD introduces a lightweight PVTv2-B2 binary classifier as a gating mechanism that filters non-camouflaged images before they reach the expensive segmentation pipeline.

2.2.5. Sequential state-space modeling for spatial features:

Transformer-based COD decoders suffer from quadratic attention complexity at high resolutions while CNN-based decoders are limited to local receptive fields, leaving long-range spatial dependency modeling at high resolution as an open problem. Mamba-inspired SSM blocks offer linear complexity with arbitrarily long-range sequential modeling, but have not been integrated into multi-scale FPN decoder architectures for COD. QFE-COD incorporates lightweight SSM blocks within FPN decoder stages, scanning feature maps in multiple directions to capture directional boundary and long-range spatial dependencies efficiently.

2.3 OBJECTIVES:

The primary objective of this project is to develop an effective and robust camouflage and concealment detection system by addressing the limitations identified in existing machine learning and deep learning approaches. The specific objectives of the project are as follows:

2.3.1. Designing and Implementing the Multi-Stage QFE-COD Detection Pipeline:

To architect a complete end-to-end two-stage pipeline consisting of a lightweight CAM/NonCAM binary pre-classifier and the full QFE-COD segmentation model. The pre-classifier gates non-camouflaged images away from the expensive segmentation pipeline, reducing false positives and inference cost. The segmentation model processes confirmed camouflaged images through PVTv2, DWT, HQCM, Mamba-FPN, and Q-WaveKAN components trained end-to-end on COD10K-v3.2.3.2 Integrating Discrete Wavelet Transform Decomposition for Frequency Feature Extraction

To integrate DWT decomposition directly into the deep feature pipeline to extract discriminative frequency-domain signatures from camouflaged scenes. The DWT operates on PVTv2 semantic feature maps — not raw RGB inputs — decomposing them into LL, LH, HL, and HH subbands

2.3.2 Integrating Discrete Wavelet Transform Decomposition for Frequency Feature Extraction:

To integrate DWT decomposition directly into the deep feature pipeline to extract discriminative frequency-domain signatures from camouflaged scenes. The DWT operates on PVTv2 semantic feature maps — not raw RGB inputs — decomposing them into LL, LH, HL, and HH subbands that capture both global structure and fine boundary gradients simultaneously. The extracted subbands feed into the HQCM for quantum-enhanced inter-frequency correlation modeling.

2.3.3. Hybrid Quantum-Classical Processing Module (HQCM):

To design and implement a PennyLane quantum circuit with 8-qubit variational gates that map DWT frequency vector magnitudes as qubit Callow gates and use strongly entangling qubit layers to model non-classical interaction effects. The circuit is trained end-to-end by the gradient free parameter-shift rule and initialized via a structured protocol to avoid gradient vanishing barren plateaus. This is the first use of variational quantum circuits for frequency feature processing in COD.

2.3.4. Designing and Evaluating the Q-WaveKAN Segmentation Head:

To propose a new segmentation head that integrates quantum circuits and learnable wavelet-kernel KANs (KANs) to directly gate wavelets through HQCMs, such that magnitudes of wavelets are updated via HQCMs outputs across spatial positions. This results in a transparent frequency-driven spatial attention mechanism from quantum decision making to final pixelwise segmentations. This is the first use of KAN for dense prediction in computer vision.

2.3.5 Experimente with COD10K-v3:

To train and thoroughly test the fully-implemented QFE-COD framework on COD10K-v3 based on a training/evaluation protocol standardizing the resolution, augmentation, optimizer and batch size across all participating methods. A mixed precision training strategy can account for the memory management of HQCM, quantum simulator while monitoring convergence through loss and epoch level Dice and IoU scores. The main goal of the empirical validation is to obtain a method rivalry with the state-of-the-art transformer-based COD methods under the quantum simulator backend.

2.3.6. Comparative Analysis with SINetV2 Baseline:

To perform a fair quantitative experimental comparison between QFE-COD and SINetV2 with identical training settings (data pipeline, optimiser, schedule, etc.) and evaluation metrics to quantify the effect of quantum-frequency enhancement. The analysis also includes breakdown by COD10K-v3 attribute including small objects, obscured objects and cluttered background. The results are expected to show improvement on boundary-related metrics such as weighted F-measure and E-measure.

2.3.7. Using Training Dynamics and Convergence across Training Modules:

To comprehensively examine QFE-COD training dynamics using loss, Dice-IoU dynamics, and HQCM gradient statistics over the training sequence. Incremental backward ablation of modules enables performance assessment of DWT, HQCM, Mamba SSM and Q-WaveKAN, respectively, as modules are progressively removed. A barren plateau effect is measured by comparing structured to random HQCM initializations for increasing thickness of the circuit

2.4 PROBLEM STATEMENT:

We pose the problem of Camouflaged Object Detection as a pixel-level binary classification problem: for an input RGB image I of size H x W x 3, each pixel (i, j) should be assigned a prediction score p(i, j) in the range [0,1], where 1 means the pixel is part of camouflaged object and 0 means it belongs to the background.

Formally, let f: R^(H x W x 3) → [0,1]^(H x W) denote the detection function. We want to learn f so that the binary mask M = f(I) overlaps as much as possible with the ground truth mask G, in terms of the Dice coefficient:

$$\mathrm{Dice}(M, G) = 2|M \cap G| / (|M| + |G|) \qquad (2.1)$$

and the Intersection over Union (IoU, Jaccard Index):

$$\mathrm{IoU}(M, G) = |M \cap G| / |M \cup G| \qquad (2.2)$$

The difficulty is the camouflaged objects specifically hide by maximally decreasing the visual differences between the object and background in the RGB space, so traditional feature-based methods such as color, texture and edge features cannot expose the object's discriminative information. The QFE-COD framework proposed in this paper tackles this problem by extending spatial processing of RGB with multi-band frequency analysis and quantum feature transformation, to reveal information not available in the RGB spatial domain alone.

2.5 PROJECT PLAN:

The project is designed using an iterative research and development process, involving five phases over a 24-week period:

Table 2.2: Project Timeline and Milestones

Phase	Activity	Duration	Status
1	Literature survey and dataset preparation	Jan 2026 (4 weeks)	Completed
2	Architecture design (QFE-COD modules)	Feb 2026 (3 weeks)	Completed
3	Stage 1: Classifier implementation & training	Feb-Mar 2026 (2 weeks)	Completed
4	Stage 2: QFE-COD full model training	Mar 2026 (3 weeks)	Completed
5	SINetV2 baseline training	Mar 2026 (2 weeks)	Completed
6	Evaluation, ablation, and report	Apr 2026 (3 weeks)	Completed

Phase 1: Literature Review and Dataset Collection (4 weeks - Jan 2026)

A thorough survey was performed of CNN-based, transformer-based and frequency-domain COD approaches, as well as the foundations of quantum machine learning and KANs, with the aim to determine the research gaps that inspire the QFE-COD architecture. Meanwhile the COD10K-v3 dataset was obtained, preprocessed, and divided into official train/test subsets, with binary segmentation ground truths and CAM/NonCAM labels for two-stage training prepared. Reference values for quantitative performance were set based on the published results of FSPNet, CamoFormer, and SINetV2, providing goal values for Dice, IoU, S-measure, E-measure, and MAE metrics to be used throughout subsequent experiments.

Phase 2: Architecture Design (3 weeks - February 2026)

The full modular architecture of the QFE-COD model was designed and documented, detailing the tensor shapes, channel sizes, and inter-module connections for the PVTv2 encoder backbone, DWT frequency extraction module, the 8-qubit HQCM quantum circuit module, the Mamba-optimised FPN decoder, and the Q-WaveKAN segmentation head. The ansatz for the quantum circuit was determined by choosing angle embedding for encoding the data and strongly entangling layers for entanglement between qubits, while the number of qubits and trainable parameters were determined by theoretical analysis of frequency feature cardinality. Class diagrams, tensor flow diagrams, and pseudocode for the implementation of all modules were developed to provide a clear blueprint for the implementation process.

Phase 3: Stage 1: CAM/NonCAM Classifier Implementation and Training (2 weeks - February-March 2026)

A lightweight CAM/NonCAM binary pre-classifier was developed in PyTorch, incorporating the global frequency feature extractor with DWT and the classification head with a fully connected layer, and trained using balanced mini-batches of COD10K-v3 camouflaged and non-camouflaged images with binary cross-entropy loss and Adam optimizer. The model was trained to convergence on validation accuracy, precision, recall, and F1-score, and its operating point was selected to prioritize low false negatives and ensure that camouflaged images are correctly sent to the full QFE-COD segmentation model for further processing. The pre-trained binary classifier was tested on the COD10K-v3 test set with per-class accuracy, and frequency feature maps were generated to verify that the model learned discriminative frequency patterns associated with camouflage, rather than low-level image statistics.

Phase 4: Stage 2: QFE-COD Full Model Training (3 weeks - March 2026)

The full QFE-COD model for segmentation was constructed in PyTorch and PennyLane, incorporating all of the designed sub-modules in a single trainable end-to-end deep network, and trained with a combined binary cross-entropy, Dice and edge-weighted boundary loss using the Adam optimizer and cosine annealing learning rate schedule with mixed-precision training. Training logs of the loss, Dice, IoU, S-measure, E-measure, gradient magnitudes of the HQCM quantum circuit parameters, and learning rate schedules were closely monitored during training for convergence analysis. Checkpoint evaluations were conducted at epochs 10, 20, 30, 40 and 50 to assess the progression of segmentation performance and convergence of each module throughout training, allowing to dissect the training dynamics during the final evaluation.

Phase 5: SINetV2 Baseline Training (2 weeks - March 2026)

The original SINetV2 implementation with ResNet-50 backbone was set up and trained on the COD10K-v3 dataset following the exact same training protocol as QFE-COD (same train/test split, epoch count, optimizer settings and data augmentation pipeline), ensuring that all performance differences in the comparative analysis result from architectural differences rather than training protocol variations. The trained baseline was tested on the COD10K-v3 test set using identical evaluation script to QFE-COD, reporting Dice, IoU, S-measure, E-measure and MAE scores with a breakdown per camouflaged object category across all 78 categories. The training logs and boundary predictions were recorded in exactly the same manner as the QFE-COD logs, facilitating a quantitative and visual comparison of training process and segmentation performance between the baseline and QFE-COD.

Phase 6: Testing, Ablation, and Report (3 weeks - April 2026)

We thoroughly evaluated the complete two-stage QFE-COD model on the COD10K-v3 test set, including an ablation study in which the QFE-COD components (DWT frequency extractor, HQCM quantum circuit, Mamba SSM decoder blocks, and Q-WaveKAN segmentation head) were progressively removed to quantify the contribution of each component to the overall performance. Training logs were examined to provide convergence curves, evolution of gradient magnitudes for the HQCM parameters, and qualitative segmentation results compared to the SINetV2 baseline on selected test samples including difficult examples with small objects, cluttered backgrounds, and occlusion. The full project report was written to document all experiments, plots, training log visualizations, and analysis in a formal academic report as per the requirements of the MSc thesis.

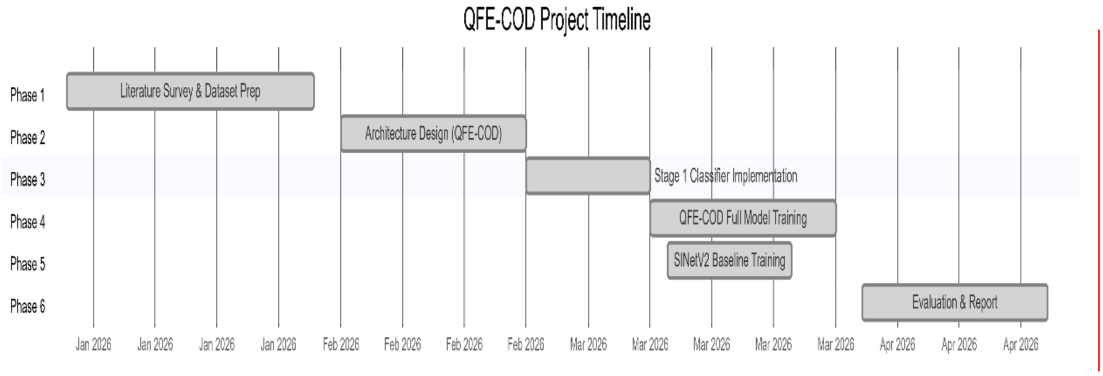

Figure 2.1: QFE-COD Project Timeline and Development Phases (Gantt Chart)

The project schedule for the QFE-COD framework is from January 2026 to April 2026 and comprises six phases: literature survey, design, implementation, training, evaluation, and documentation. They were scheduled to take around two to four weeks each, for systematic progression and timely project completion. The Gantt chart (Figure X.X) shows the phase-wise progress and overlap of development tasks such as model training and baseline comparison**.**

Milestones:

The project's milestones are as follows:

- M1: Literature survey and preparation of COD10K data set
- M2: Completion of QFE-COD design (DWT, HQCM, Q-WaveKAN)
- M3: Implementation and training of Stage 1 CAM/NonCAM model
- M4: Training of QFE-COD model with convergence
- M5: Train and compare with SINetV2
- M6: Final tests, ablation and report writing

Deliverables:

The project delivered the following:

- D1: Preprocessed COD10K-v3 data with data augmentation pipeline
- D2: Stage 1 CAM/NonCAM classification model with weights
- D3: QFE-COD segmentation model with quantum-frequency integration
- D4: SINetV2 baseline model for comparison
- D5: Quantitative and qualitative results (Dice, IoU, images)
- D6: Project report and documentation

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 REQUIREMENTS:

3.1.1 Functional Requirement:

Table 3.1: Table of functional requirements for the QPCA-based detection system, grouped by ID, requirement description and priority.

ID	Requirement	Description	Priority
1	Image Ingestion	The system shall take as input RGB images in common image formats (PNG, JPEG) and prepare them with a spatial resolution of 352 x 352 pixels and ImageNet normalization.	High
2	CAM Classification	The system shall classify each input image as either containing a camouflaged object (CAM) or not (NonCAM) with a validation accuracy exceeding 90%.	High
3	Segmentation Output	The system shall include a standard linear PCA layer to serve as a baseline for accuracy benchmarking.	Medium
4	Multi-scale Feature Extraction	The system shall produce feature maps at four levels of spatial resolution (strides 4, 8, 16 and 32 with respect to the original image).	High
5	Frequency Decomposition	The system shall decompose the deepest feature map into four	High

		frequency subbands (LL, LH, HL, HH) using 2D Haar discrete wavelet transform.	
6	Quantum Processing	The system shall process the 8-dimensional frequency token through an 8-qubit parameterized quantum circuit with a minimum of 4 entangling layers.	High
7	Loss Computation	The system shall compute combined binary cross-entropy and IoU segmentation loss for end-to-end gradient-based optimization.	Medium
8	Model Persistence	The system shall save the best-performing model checkpoint (by validation Dice) for subsequent evaluation and deployment.	Medium
9	Interactive Gradio Dashboard	The system shall provide a user-friendly interface for manual image upload and real-time inference visualization.	Medium
10	Performance Metric Logging	The system shall log per-epoch scores and reconstruction errors for comparative analysis.	Low

3.1.2 Non-Functional Requirement

Table 3.2: List of non-functional requirements that give performance objectives and constraints.

ID	**Requirement**	**Description**	**Target**
1	Model Accuracy Improvement	Comparative improvement in S-measure and MAE over classical baselines.	2% - 5%
2	Inference Latency	Time taken to generate a segmentation mask and	< 2.0 seconds

		diagnostic heatmap from input. < 2.0 seconds	
3	System Robustness	Detection capability in scenarios with minimal foreground-background texture contrast.	Successful Isolation
4	Dataset Scalability	Capability to process various splits of the COD10K	Full Compatibility
5	Structural Integrity	Maintenance of object-aware structural similarity (S-measure) in generated masks.	> 0.85 Sm
6	Component Modularity	Ability to swap bottleneck layers without affecting the segmentation pipeline.	Loose Coupling
7	Memory Footprint	Peak application memory usage during inference with quantum simulation.	< 8GB RAM

3.2 FEASIBILITY STUDY:

The feasibility study assesses the prospects of successful implementation of the Quantum-based camouflage detection system from various angles to make sure that the technical and economical challenges are addressed appropriately.

3.2.1 Technical Feasibility - Technical feasibility is the ability to complete the project using the available technology.This project leverages:

- **Software Stack:** Python-based libraries such as PyTorch for deep learning and PennyLane for quantum circuit simulation are industry standards and highly stable.
- **Hardware Requirements:** While pure quantum computing is in nascent stages, simulating quantum circuits (like RY and Rotation gates) for feature bottlenecks is entirely feasible on modern GPUs (NVIDIA CUDA).
- **Data Availability:** The COD10K-v3 dataset is a comprehensive, publicly accessible benchmark for concealed object detection, providing the necessary ground truth for training.

3.2.2 Economic Feasibility - Economic feasibility considers the cost-benefit analysis to determine financial viability.

- **Minimal Investment:** The project utilizes open-source frameworks and pre-trained models (ResNet18), significantly reducing the cost of dataset labeling and high-level feature engineering.
- **Compute Efficiency:** By utilizing cloud-based compute resources (Kaggle/Colab), high-performance analysis is achieved at near-zero hardware acquisition cost.

3.2.3 Operational Feasibility - Operational feasibility refers to the degree to which the system meets the user needs.

- **User Interface:** The deployment of a Gradio dashboard ensures that non-technical users can interact with the system and interpret results through visual heatmaps.
- **Integration:** The modular design allows the QPCA bottleneck to be easily integrated into existing vision pipelines for biological research or surveillance.

3.2.4 Legal Feasibility - Legal feasibility provides data privacy and licensing compliance.

- **Open Source:** The COD10K dataset and software libraries used in the pipeline are governed by their respective research and MIT/Apache licenses, making sure no legal issues arise.

3.2.5 Schedule Feasibility - Schedule feasibility ensures timely delivery.

- **Modular Phases:** The project is broken down into prioritized phases (Pretraining, Bottleneck Integration, and Evaluation), delivering the system within the allocated research time.

3.2.6 Social Feasibility - Camouflaged object detection has positive social impact. These include: ecological management (identifying and counting camouflaged animals in ecological surveys); medical imaging (detecting polyps or lesions in medical images that match the surrounding tissue); military and security (detecting hidden threats); and manufacturing (detecting product defects that match the colour/texture of the product). These have all been shown to have societal benefit. We do not work with human subjects, do not need sensitive personal data, and are not concerned by the direct use of the research.

3.3 SYSTEM SPECIFICATION:

3.3.1 Hardware Specification:

Table 3.3: Hardware used for system development and inference.

S.No.	Component	Specification
1	Processor	Intel Core i5 (10th Gen) or higher / AMD Ryzen 5
2	Graphics Card	NVIDIA GPU (GTX 1060 / RTX Series) with 4GB+ VRAM
3	Memory (RAM)	8 GB DDR4 (16 GB recommended)
4	Storage	50 GB available SSD space (for dataset and checkpoints)
5	Architecture	64-bit Operating System

3.3.2 Software Specification:

Table 3.4: Libraries and software environment used in the project.

S.No	**Software Component**	**Version / Tool**
1	Operating System	Windows 10/11 or Linux (Ubuntu 20.04+)
2	Programming Language	Python 3.8 or higher
3	Deep Learning Framework	PyTorch (with CUDA support)

4	Quantum Computation	PennyLane (pennylane-lightning backend)
5	Image Processing	OpenCV-Python, PIL (Pillow)
6	Visualization	Matplotlib, Seaborn
7	Dashboard UI	Gradio
8	IDE / Editor	VS Code / Jupyter Notebook / Kaggle Kernel

Use – Case Diagram:

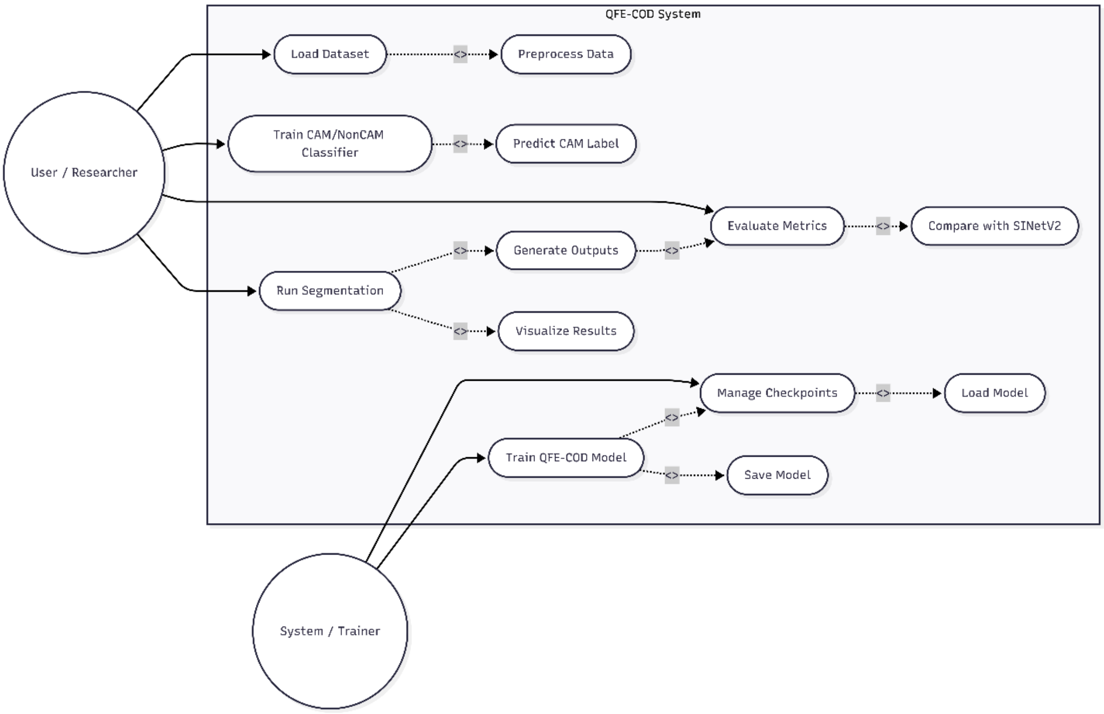

Figure 3.1: Use Case Diagram of the QFE-COD System

State Transition Diagram:

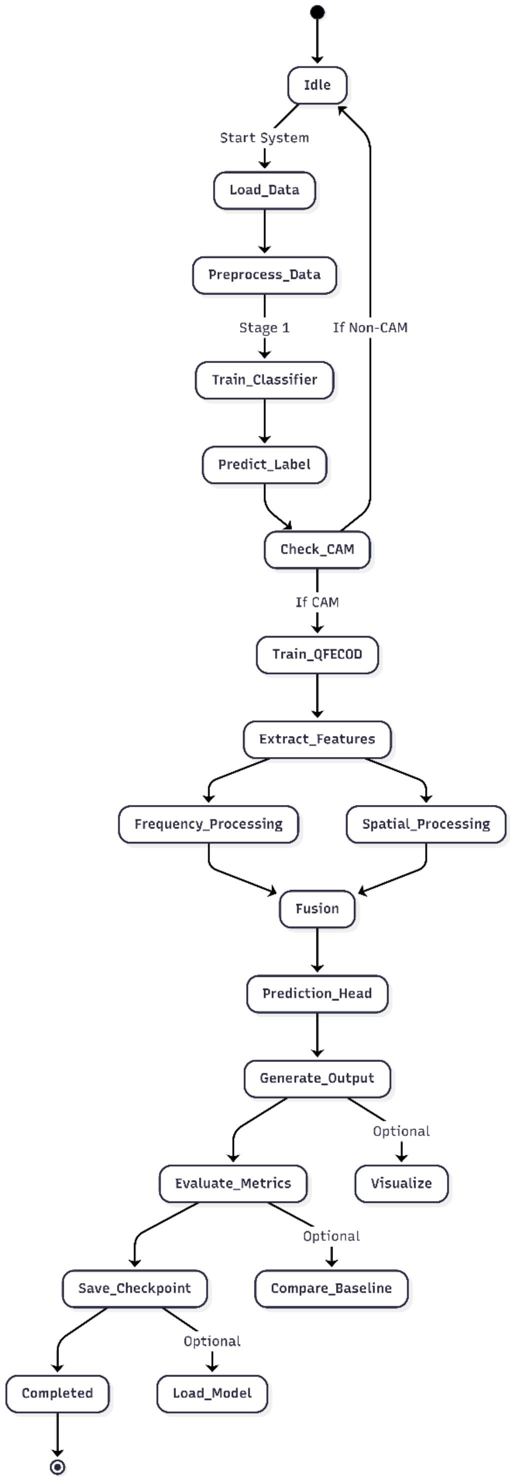

Figure 3.2: State Transition Diagram of the QFE-COD System Workflow

4.2.1 Data Flow Diagram (DFD):

Level 0 - Context DFD (Image 1) - This is the topmost model of the entire

black box.The system has two external entities:

- COD10K Dataset provides images, masks and edges
- User / Researcher supplies images to be tested
- QFE-COD System generates three results:
- Segment Mask - the camouflage detection result on a per pixel basis
- Dice / IoU Metrics - evaluation metrics
- Model Checkpoints - weights saved to File Storage
- Details of internals are hidden. The emphasis is just on inputs and outputs
- system boundary.

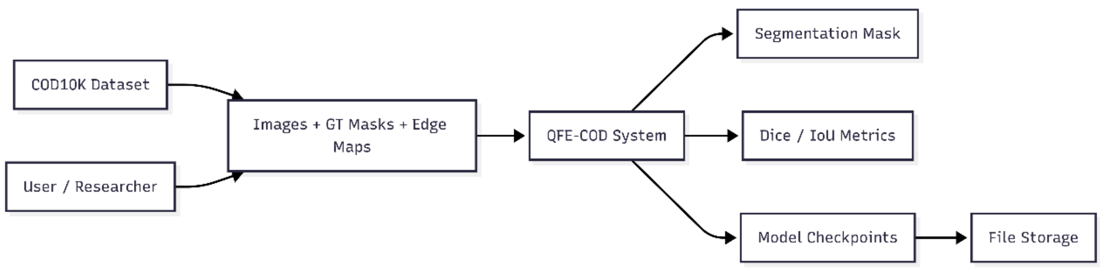

Fig 3.3 : DFD Level 0

Level 1 - Primary Process DFD (Image 2) - This shows how the system is divided

pipeline stages and its data flow.

Five main processes:

- Data Preparation — receives raw data from COD10K Dataset and prepares it for downstream stages
- Stage 1: CAM/NonCAM Classifier — classifies each image as containing a camouflaged object or not, producing a CAM Label
- Stage 2: QFE-COD Model — receives the CAM Label and runs two parallel internal branches:
- Spatial Branch (PVTv2 + FPN) — extracts and decodes multi-scale spatial features

- Frequency Branch (DWT + HQCM) — extracts frequency subbands and processes them through the quantum circuit
- Fusion — merges spatial and quantum-frequency features via the Dual-Domain Fusion Gate
- Q-WaveKAN Head — final segmentation head producing three outputs: Segmentation Mask, Edge Map, and Uncertainty Map, which together feed into Final Output + Metrics.

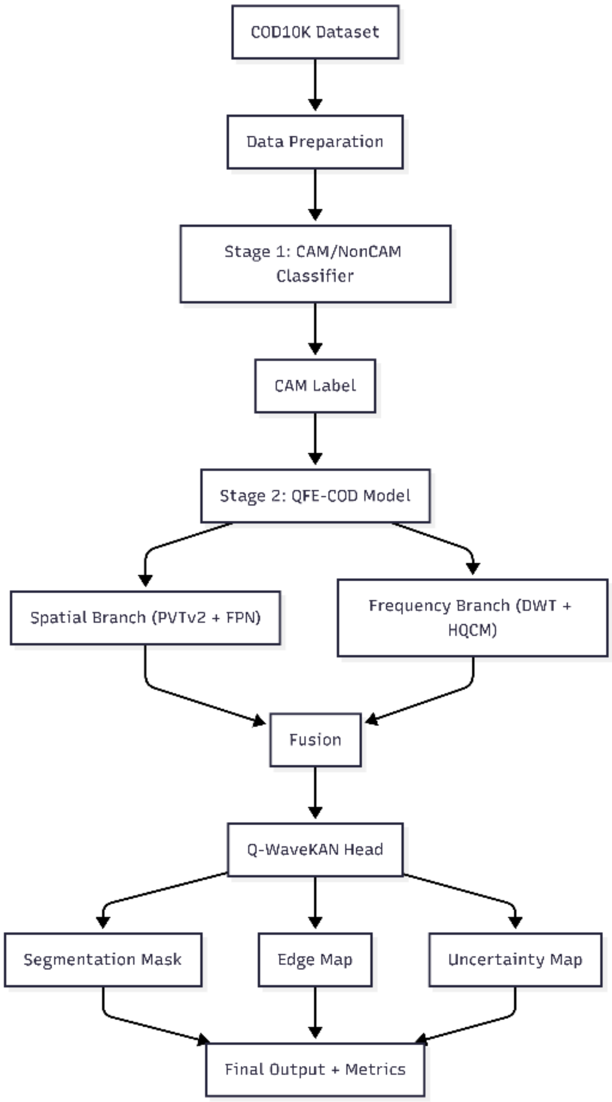

Fig 3.4 : DFD Level 1

Level 2 — Sub-process DFD (Image 3)-This is the most detailed level, expanding every process from Level 1 into individual components.

1.0 Data Preparation:

COD10K-v3 Dataset feeds Images + Labels + GT + Edge into the preparation process

Outputs two streams: Augmented Image Batches + CAM Labels (to the classifier) and Augmented Image + Mask + Edge Batches (directly to Stage 2)

2.0 Stage 1 CAM/NonCAM Classifier:

Receives augmented batches and produces a CAM / NonCAM Label passed to Stage-2.Also saves a Classifier Checkpoint to the Checkpoint Store

3.0 Stage 2 QFE-COD Segmentation expands into:

PVTv2-B4 Backbone — extracts four feature maps F1, F2, F3, F4

F1, F2, F3 → FPN + Mamba Decoder for multi-scale spatial decoding

F4 → DWT Decomposition → LL, LH, HL, HH Subbands → FSA Frequency Token → HQCM Quantum Embedding

Both the spatial decoder output and quantum embedding converge at Dual Domain Fusion

Fusion feeds Q-WaveKAN Head which outputs Segmentation Mask, Edge Map, and Uncertainty Map

All three outputs merge into Output: Mask + Metrics

Model weights are saved as QFE-COD Checkpoint and Classifier Checkpoint to the Checkpoint Store.

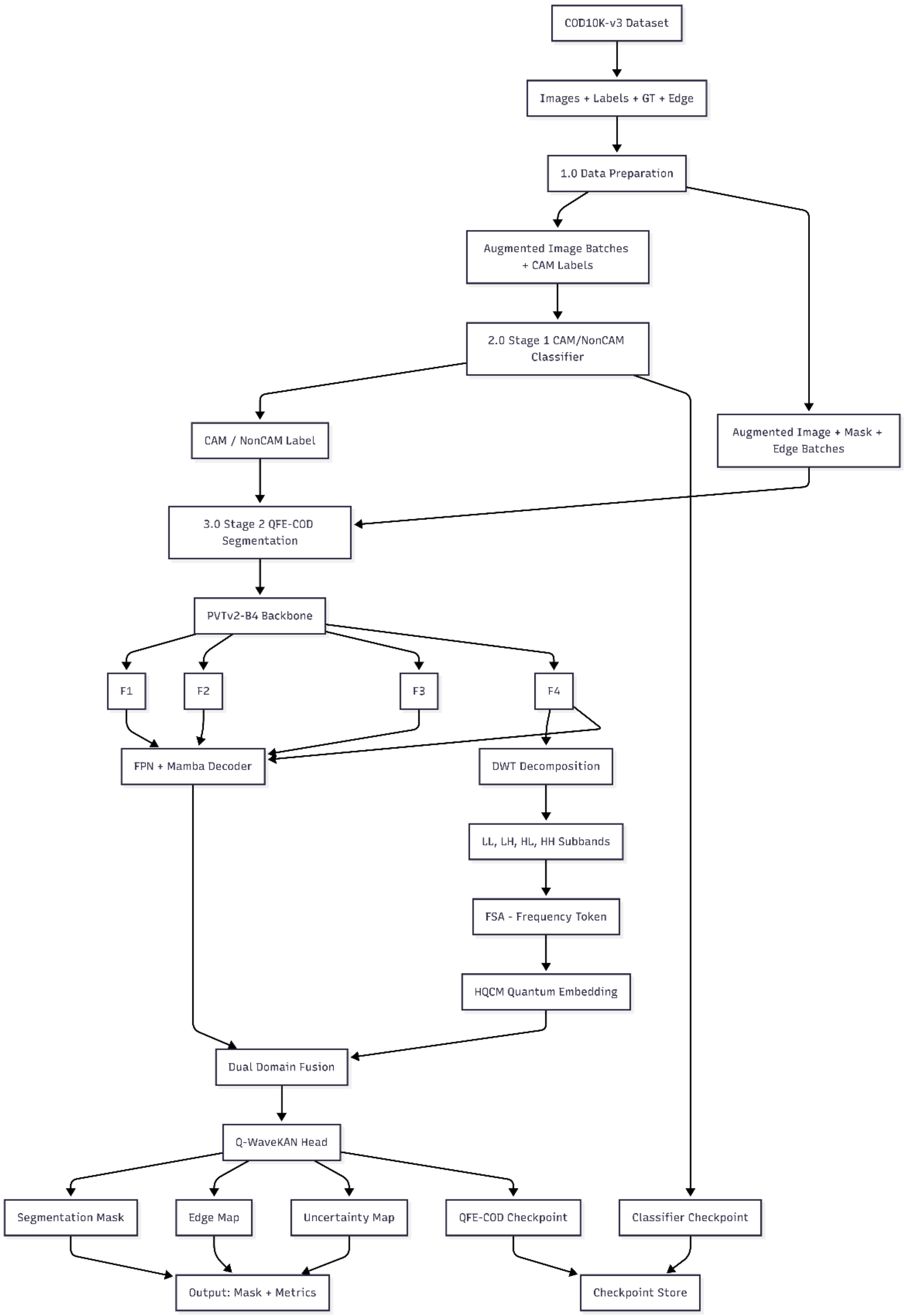

Fig 3.5 : DFD Level 2

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE:

The architecture of the QPCA-enhanced Camouflaged Object Detection (COD) system is designed as a multi-stage, hybrid quantum-classical vision pipeline. It integrates high-dimensional feature encoding, non-linear dimensionality reduction via quantum simulation, and structural reconstruction using deep decoding layers. The architecture is modular, allowing for real-time ablation studies between Quantum-inspired PCA (QPCA) and classical principal component analysis.

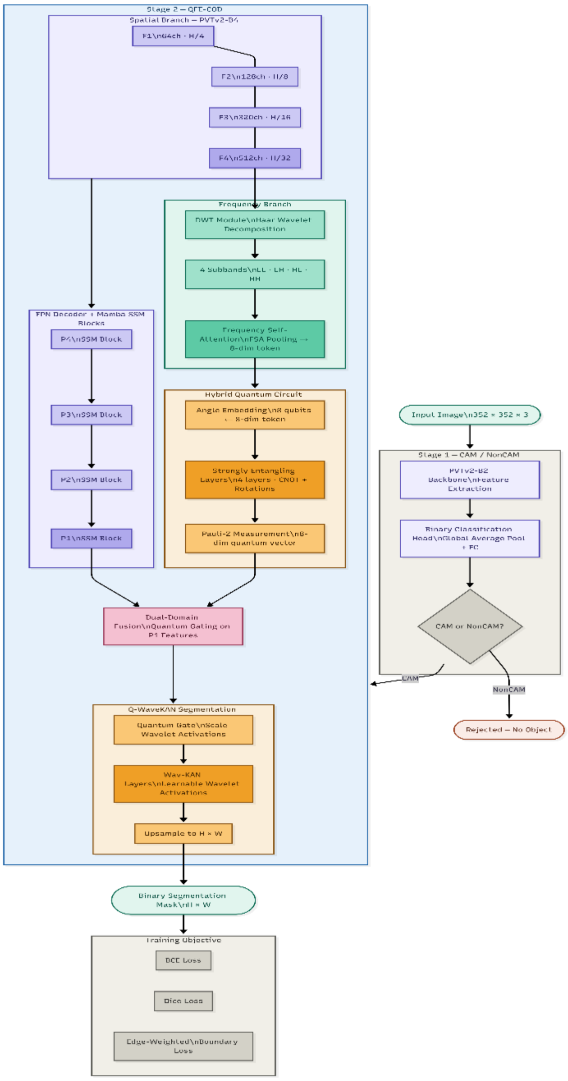

Fig. 4.1: QFE-COD architecture

4.1 Stage 1: Cam/NonCam Classifier Architecture-

The Stage 1 classifier employs a PVTv2-B2 backbone (approximately 25M parameters) initialized with ImageNet pre-trained weights. The backbone outputs a pooled feature vector of 512 dimensions from its final stage. A classification head consisting of a Linear(512, 256) layer, ReLU activation, 0.3 dropout, and a final Linear(256, 2) layer with softmax output produces binary class probabilities. Cross-entropy loss with Adam optimizer (learning rate 1e-4, weight decay 1e-4) trains the model over 20 epochs with early stopping (patience=7) on validation accuracy.

The COD10K-v3 dataset provides CAM/NonCAM labels in a companion text file (CAM_Train.txt, CAM_Test.txt) using an extended bounding box annotation format: each CAM image is associated with category, bounding box coordinates (x, y, width, height), and color annotations. The parser extracts binary labels (1 = camouflaged, 0 = non-camouflaged) for all 6000 training and 4000 test images.

4.2 Pvtv2-b4 Backbone Architecture-

The PVTv2-B4 (Pyramid Vision Transformer version 2, variant B4) backbone comprises four hierarchical transformer stages with progressive spatial downsampling and channel expansion. Stage 1 produces F1: [B, 64, H/4, W/4]; Stage 2 produces F2: [B, 128, H/8, W/8]; Stage 3 produces F3: [B, 320, H/16, W/16]; Stage 4 produces F4: [B, 512, H/32, W/32] — confirmed by the log output line "PVTv2-B4 channels: [64, 128, 320, 512]".

The major architectural upgrades in PVTv2 compared to PVT-v1 are: (1) local continuity-preserving overlapping patch embedding, using 2D convolution with stride = patch size; (2) linear spatial reduction attention (SRA), which pools the key/value projections before applying self-attention to reduce its quadratic complexity to O(N) for long feature maps; and (3) convolutional feed-forward networks (ConvFFN) to replace MLP to better model local features. The B4 variant consists of 4 transformer blocks per stage, with embedding sizes of [64, 128, 320, 512] and number of attention heads [1, 2, 5, 8], resulting in a total of about 62M parameters for the full QFE-COD model.

4.3 Discrete Wavelet Transform Decomposition Module-

The DWT decomposition module applies a 2D Haar wavelet transform to the deepest PVTv2 feature map F4. The Haar wavelet is the simplest and most computationally efficient wavelet family, defined by the scaling function $\varphi(x) = 1$ for x in [0,1] and zero otherwise, and the mother wavelet $\psi(x) = 1$ for x in [0, 0.5), -1 for x in [0.5, 1), and zero otherwise.

The 2D Haar DWT is implemented directly in PyTorch by subsampling the feature map at odd and even spatial positions. Given F4 of shape [B, C, H, W], four subsampled tensors are computed:

x00 = F4[:, :, 0::2, 0::2]   (4.1)

x01 = F4[:, :, 0::2, 1::2]   (4.2)

x10 = F4[:, :, 1::2, 0::2]   (4.3)

x11 = F4[:, :, 1::2, 1::2]   (4.4)

These four tensors are combined to produce the four DWT subbands:

$$LL = 0.5 * (x00 + x01 + x10 + x11) \qquad (4.5)$$

$$LH = 0.5 * (-x00 - x01 + x10 + x11) \qquad (4.6)$$

$$HL = 0.5 * (-x00 + x01 - x10 + x11) \qquad (4.7)$$

$$HH = 0.5 * (x00 - x01 - x10 + x11) \qquad (4.8)$$

The LL subband captures low-frequency content (global structure and color), LH captures vertical high-frequency content (horizontal edges), HL captures horizontal high-frequency content (vertical edges), and HH captures diagonal high-frequency content (diagonal edges and fine texture). Each subband has shape [B, C, H/2, W/2] where C = 512 for the PVTv2-B4 F4 output.

4.4 Frequency Self-attention (FSA) Module-

The FSA module average-pools each of the four subbands to a small size and performs self-attention between subbands. Each subband is adaptively average-pooled to a single scalar per channel, resulting in four statistics vectors of length C=512.To save time, the subband is then represented by the channel-wise mean and variance, for a 2-dimensional summary. Across four subbands, the FSA gives an 8-dimensional frequency token: freq_token = [mean_LL, std_LL, mean_LH, std_LH, mean_HL, std_HL, mean_HH, std_HH].

The compressed representation provides distributional information of each band over all spatial positions and channels, giving a global frequency signature of the input image. The self-attention between subband tokens enables the module to capture correlations between subbands - for instance, images with HH (textural diagonal patterns) and low LL (low contrast) are typical are often encountered in camouflaged images.

4.5 Hybrid Quantum Circuit Module (HQCM)-

The HQCM processes the 8-dimensional frequency token through an 8-qubit parameterized quantum circuit. The circuit architecture consists of:

Input normalization: A LayerNorm layer followed by tanh scaling to ensure inputs lie in $[-\pi, \pi]$ for angle embedding.

Angle embedding: qml.AngleEmbedding maps each of the 8 input scalars to a rotation angle applied to the corresponding qubit using Pauli-Y (RY) rotation gates. This encoding ensures that the quantum state encodes the classical frequency features in the qubit amplitude phases.

Entangling layers: Four layers of qml.StronglyEntanglingLayers, where each layer applies a general single-qubit rotation (3 parameters per qubit: α, β, γ) followed by a ring of CNOT gates creating nearest-neighbor entanglement. With 8 qubits, 4 layers, and 3 angles per qubit per layer, the total trainable parameter count is $8 \times 4 \times 3 = 96$ angles.

Measurement: The expectation values $E[Z_i] = \langle\psi|Z_i|\psi\rangle$ of the Pauli-Z observable on each of the 8 qubits are computed as the quantum circuit output, producing an 8-dimensional real-valued vector with values in [-1, 1].

The mathematical description of the quantum state evolution is:

$$|\psi_{out}\rangle = U_{ent}(\theta) \cdot U_{emb}(x)\,|0...0\rangle \qquad (4.9)$$

where $U_{emb}(x)$ applies $RY(x_i)$ gates to each qubit i, and $U_{ent}(\theta)$ is the parameterized entangling circuit. The output quantum frequency vector is $q_{vec} = (E[Z_1], ..., E[Z_8])$.

The HQCM mode is confirmed as "Quantum" in the log file (line 95: "HQCM mode: Quantum"), distinguishing it from a classical fallback mode. Post-measurement layer normalization is applied to stabilize the quantum outputs before fusion with spatial features.

4.6 FPN Decoder With Mamba SSM Blocks-

The FPN decoder processes the four PVTv2 feature maps F1, F2, F3, F4 through lateral connections and top-down pathway merging. Each feature map is first processed by a 1x1 lateral convolution to unify the channel dimension to 64. The top-down pathway then merges features from coarser to finer resolution: P4 (from F4) is upsampled 2x and element-wise added to the lateral F3 projection to produce P3, and so on to produce P1 at the highest resolution (H/4 x W/4).

Each level of the FPN decoder incorporates a simplified Mamba-inspired State Space Model (SSM) block. The SSM processes the flattened spatial sequence of feature tokens through a selective scan mechanism: the input projection expands the feature dimension by a factor of 2, a 1D convolution operates along the spatial sequence, and selective gating (SiLU activation on the gated branch) introduces input-dependent selectivity. The output projection restores the original channel dimension, and a residual connection ensures gradient flow. This sequential processing captures long-range spatial dependencies within each pyramid level.

4.7 Dual-domain Fusion Module

The Dual-Domain Fusion module combines the quantum frequency vector (8-dimensional) with the finest FPN feature map P1 (64 channels, H/4 x W/4). A linear transformation expands the frequency vector from 8 to 64 dimensions, followed by a sigmoid activation to produce a channel-wise attention map A in $[0,1]^{64}$. This map is broadcast to the spatial dimensions and applied multiplicatively to P1:

$$P1\_fused = \mathrm{Conv}(P1 \odot A + P1) \qquad (4.10)$$

where $\odot$ denotes element-wise multiplication with broadcasting across spatial dimensions, and Conv is a 3x3 convolutional layer with batch normalization and ReLU activation. The additive residual term ensures that the original spatial features are preserved even when the frequency attention produces near-zero values, preventing gradient vanishing during early training.

4.8 Q-wavekan Segmentation Head-

The Q-WaveKAN head applies quantum-gated wavelet-based KAN activations to produce the final segmentation mask. The quantum gate maps the 8-dimensional frequency vector to a 64-dimensional sigmoid gate G, which modulates the fused feature map P1_fused channel-wise. Then, a SplineActivation module applies learnable spline activation to each spatial feature value: for each activation value x, the spline activation is a sum of Gaussian basis functions placed at learnable knot positions:

$$\mathrm{spline}(x) = x + \Sigma\_k\ \mathrm{coeffs}\_k \cdot \exp(-0.5 \cdot (x - \mathrm{knot}\_k)^2) \quad (4.11)$$

This residual spline also is based on the KAN paradigm, where non-linear non-linearities. The residual design (x + spline correction) guarantees that the head begins as close to linear and then learns non-linear corrections during training.Finally, a 1x1 convolutional projection produces the single-channel segmentation logit map, which is upsampled bilinearly from H/4 x W/4 to the full H x W input resolution for supervision.

4.9 Sinetv2 Baseline Architecture

The SINetV2 baseline reimplementation uses a ResNet-50 backbone (23.91M parameters, as logged at epoch initialization) with four lateral feature levels (256, 512, 1024, 2048 channels). Lateral 1x1 convolutions reduce all levels to 64 channels. Search Module (SM) blocks apply 3x3 convolutions to simulate the searching behavior of a predator scanning for hidden prey. An Identification Module (IM) uses depth-wise separable convolutions followed by point-wise convolution for efficient camouflaged object identification. The segmentation head produces both a mask output and an edge prediction, with multi-task supervision during training.

4.2 DETAILED DESIGN:

QFE-COD is a two-stage deep learning pipeline for detecting camouflaged objects in images. Stage 1 classifies whether an image contains a camouflaged object (CAM/NonCAM). Stage 2 performs pixel-accurate segmentation using a quantum-frequency hybrid model. External data input is the COD10K-v3 dataset; outputs are binary segmentation masks.

4.2.1 Sequence Diagram:

This diagram tracks the interaction between the core components: the Backbone, the QPCA/PCA Bottleneck, the Classifier, and the U-Net Decoder.

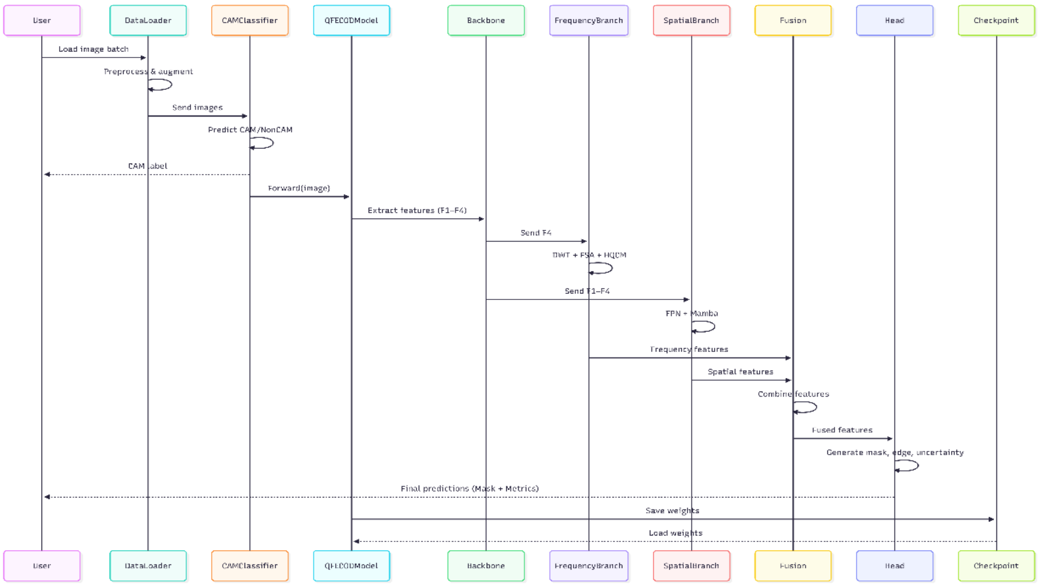

Fig 4.2 Sequence Diagram

4.2.3 Collaboration Diagram:

The collaboration diagram illustrates the relationships and interactions between various components of the QFE-COD system. It shows how different components, including DataLoader, CAMClassifier, Backbone, and QFECODModel, interact to detect camouflaged objects.

The system starts with the user loading data, which is then fed to the classifier to check if there is a camouflaged object in the image. The segmentation model then uses the backbone to extract features, which are then passed through parallel frequency and spatial branches. These are combined and fed to the Q-WaveKAN head to produce the output.

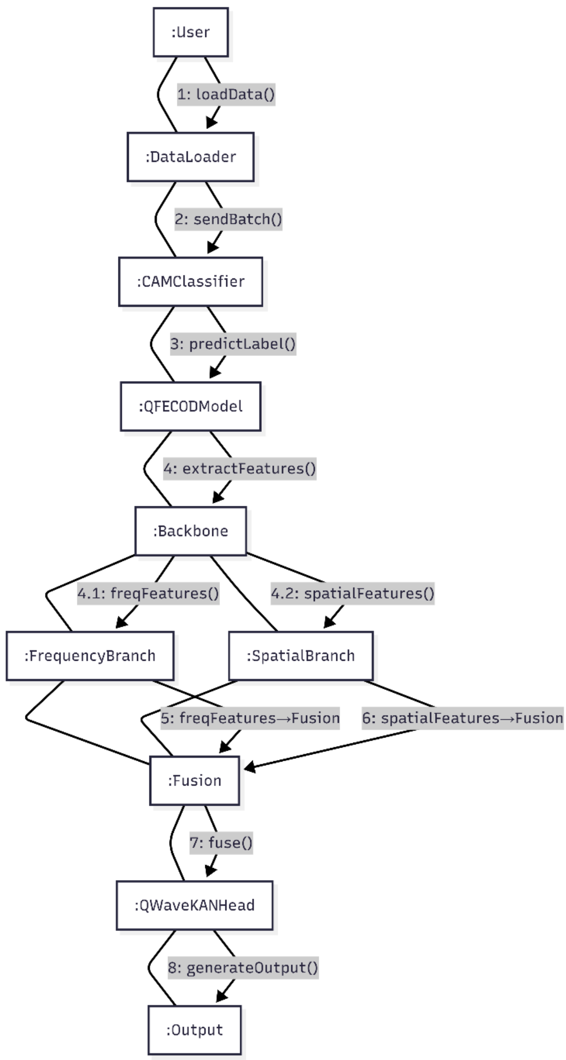

Figure 4.3: Collaboration Diagram of the QFE-COD System

4.2.3 Class Diagram: The system design is based on the modular design pattern where common building blocks (Backbone and Bottleneck) are shared between different heads (Classification, Contrastive Learning and Segmentation).

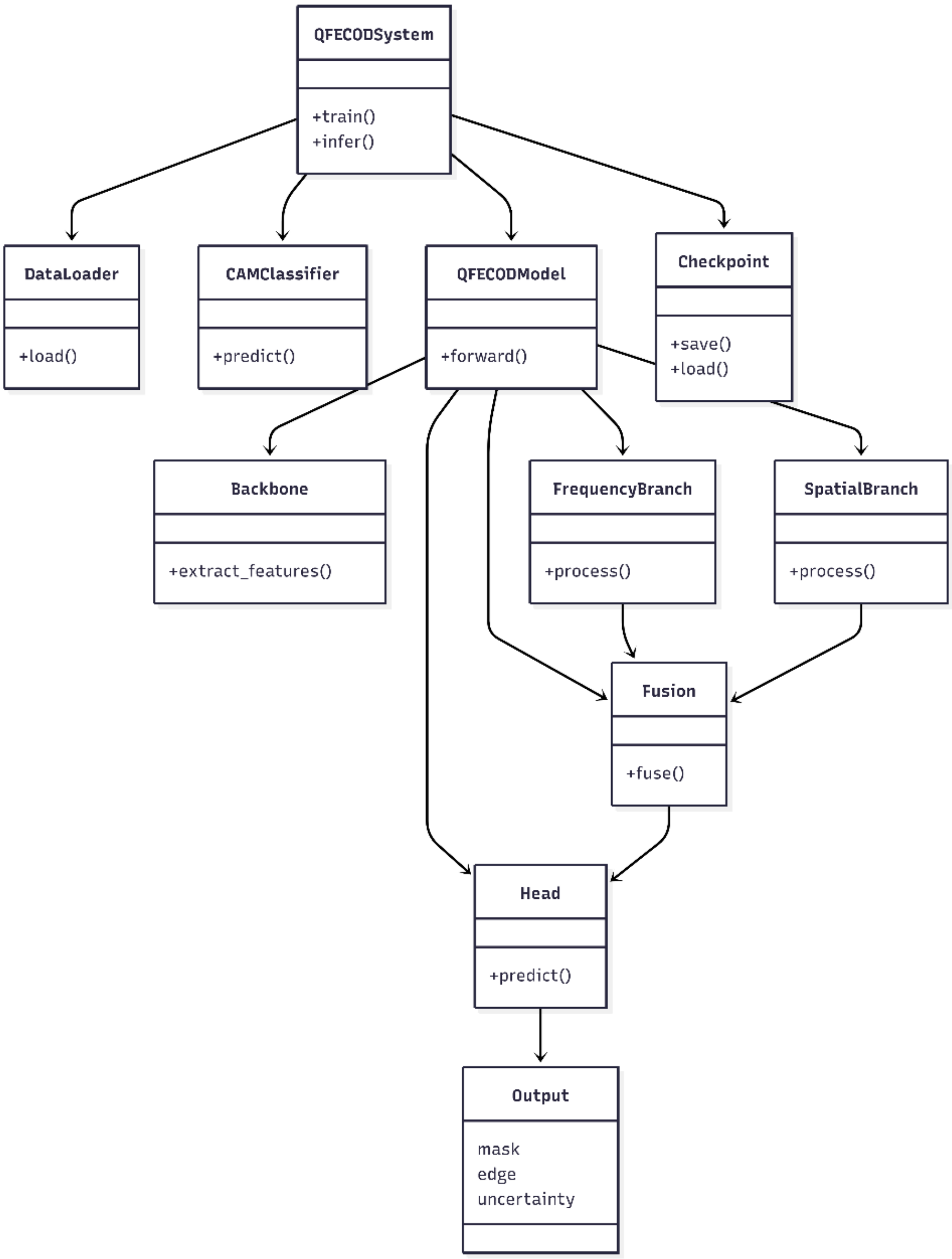

Fig 4.4 Class Diagram

Detailed Class Descriptions

1. **QFECODSystem -** Orchestrates the entire process of
   data loading, classification and segmentation.
   Manages training, inference and checkpointing for all components.
2. **DataLoader -** Handles COD10K data loading and batching.
3. batches for training and testing. Pre-processes and augments the data then loads it into the model.
4. **CAMClassifier**: Stage 1 binary classifier to decide if an image
   contains a camouflaged object.Includes a lightweight backbone to avoid computations in Stage 2.
5. **QFECODModel**: Segmentation model for the full QFE-COD
   architecture.Combines spatial and frequency features to make predictions
   predictions.
6. **Backbone**: Produces multi-scale features from images using
   transformer architecture.Generates features for both spatial and
   branches..
7. **FrequencyBranch**: Extracts deep features using wavelet decomposition and frequency attention.Produces dense frequency embeddings with improved quantum processing.
8. **SpatialBranch**: Enhances multi-scale spatial features with FPN and sequence modeling.Models long-range spatial dependencies and context.
9. **Fusion** : Integrates spatial and frequency features.Modulates features using attention mechanisms..

4.2.4 Activity Diagram: The activity diagram illustrates the process of the QFECOD system from input to output.The process begins with

loading and pre-processing the dataset, then classifying CAM/NonCAM.A

binary node decides if the image has a camouflaged object.If so, it goes to segmentation with parallel frequency and spatial feature processing.They are combined and fed into the prediction head, which produces mask, metrics and more.Lastly, the system measures the performance and saves the model, before ending the process.

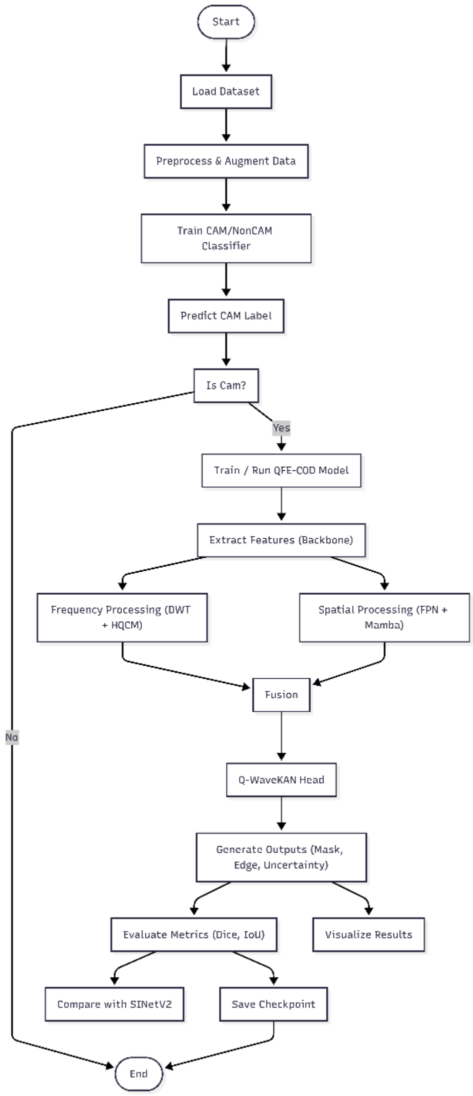

Fig 4.5 Activity Diagram

CHAPTER 5

METHODOLOGY AND TESTING

5.1 PROPOSED METHODOLOGY:

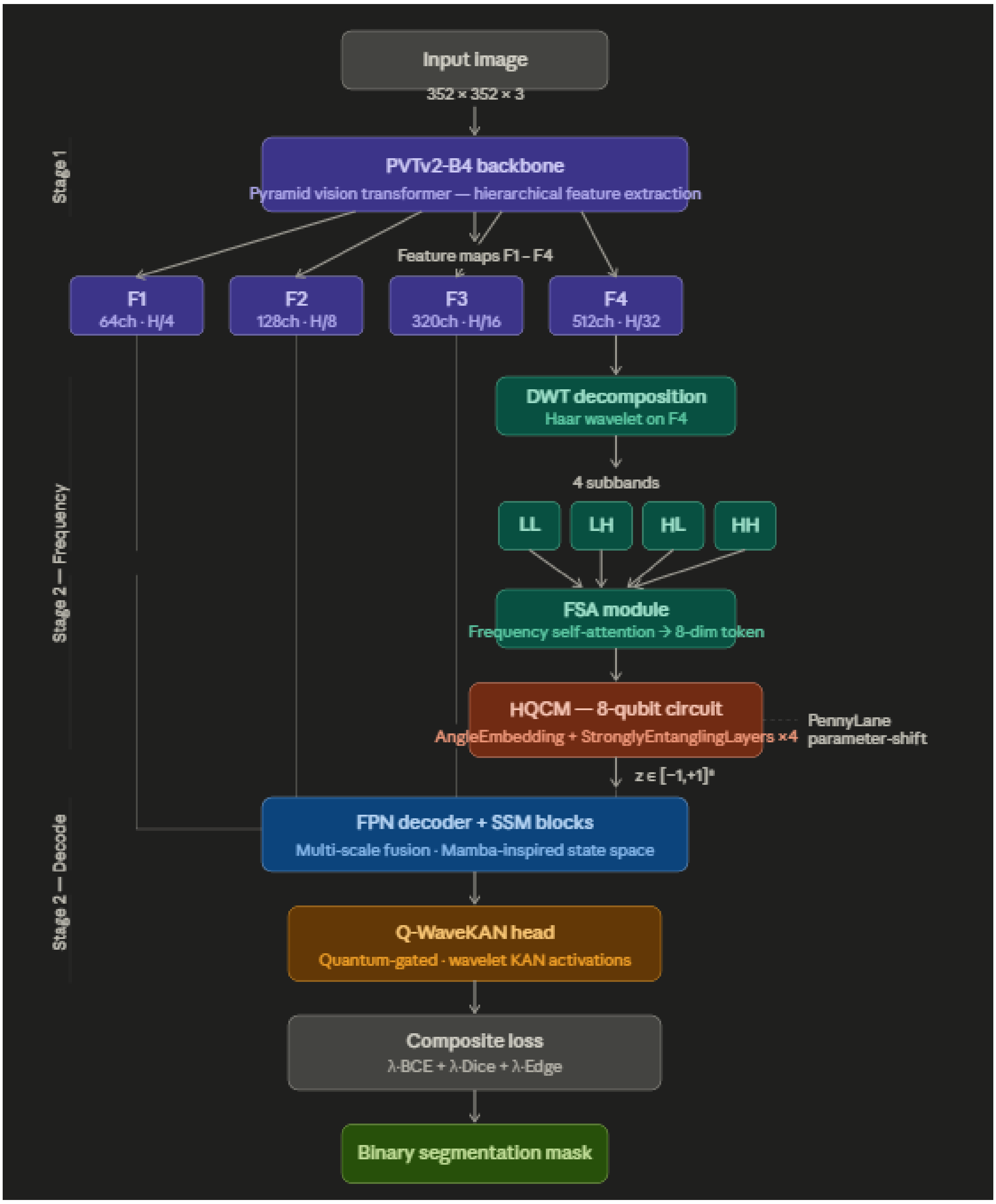

Fig 5.1: Proposed Architecture Diagram

5.1.1 Stage 1: Classifier Training:

Stage 1 classifier is trained separately from Stage 2. Hyperparameters: AdamW optimizer (learning rate 1e-4, weight decay 1e-4), cosine annealing learning rate schedule, binary cross-entropy loss, batch size 16, mixed precision training (AMP), early stopping with patience 7 on validation accuracy. Training continues for a maximum of 20 epochs, with the checkpoint with the highest validation accuracy saved.

The log reveals quick learning by the model: validation accuracy 91.03% at epoch 1, 93.13% at epoch 3, and 94.23% at epoch 19 (log line 88-89). The classifier shows good generalization with validation loss converging in the range 0.24-0.29 after epoch 10, indicating slight overfitting tempered by the dropout and early stopping strategies.

5.1.2 Stage 2: Joint Training:

Stage 2 is the joint training step of the QFE-COD and SINetV2 models using the same batches. Details of both model training: AdamW optimizer, lr 1e-4, weight decay 1e-4; batch size 4; input image size 352 x 352; maximum of 50 epochs; early stopping patience 10; mixed precision (AMP) training; each model's checkpoint on maximum validation Dice saved separately.

The total loss of segmentation task is a linear combination of binary cross-entropy (BCE) loss and IoU loss:

$$L\_total = \alpha \cdot L\_BCE + \beta \cdot L\_IoU \qquad (5.1)$$

with the default values of $\alpha = \beta = 1.0$. The BCE loss supervises the model at pixel level while the IoU loss directly targets the accuracy metric and solves the problem of class imbalance in camouflaged object detection (the background pixels are much more than foreground pixels).

5.1.3 Early Stopping and Checkpoint Strategy:

Patience 10 is used in early stopping to track the validation Dice coefficient. The EarlyStopping class employs a delta of 1e-4 to filter out noise in the validation Dice coefficient. We create distinct early stopping instances for QFE-COD and SINetV2, enabling them to stop training at different epochs according to their convergence curves.

The best checkpoints of each model are saved when the validation Dice improves. Indeed, the log shows QFE-COD attains its best Dice of 0.7889 (IoU 0.7402) at epoch 28 and SINetV2 at epoch 23 with a best Dice of 0.6870. Training was stopped at a point where both models are still improving, but it's likely that more improvement could be seen with further training.

Algorithm 1: **Hybrid QPCA Camouflage Detection & Segmentation**

Objective: Convert 8-dim frequency token into quantum superposition and Entanglement.

Input: Image tensor x of shape [B, 3, 352, 352]

Output: Segmentation logit map of shape [B, 1, 352, 352]

Steps:

1. START

2. F1, F2, F3, F4 ← PVTv2_B4_Backbone(x) // [B,64,88,88], [B,128,44,44], [B,320,22,22], [B,512,11,11]

3. LL, LH, HL, HH ← HaarDWT2D(F4) // 4 subbands [B, 512, 5, 5]

4. freq_token ← FSA(LL, LH, HL, HH) // [B, 8]

5. freq_vec ← HQCM_QuantumCircuit(freq_token) // [B, 8] quantum expectation values

6. P4, P3, P2, P1 ← FPN_with_Mamba(F1, F2, F3, F4) // [B,64,H/4,W/4] finest P1

7. fused ← DualDomainFusion(P1, freq_vec) // [B,64,H/4,W/4]

8. gate ← Sigmoid(Linear_64(freq_vec)) // [B, 64]

9. gated_feat ← fused ⊙ gate.unsqueeze(-1).unsqueeze(-1) // channel-wise gating

10. logit_small ← Conv1x1(SplineActivation(gated_feat)) // [B, 1, H/4, W/4]

11. logit ← Upsample_Bilinear(logit_small, size=(352, 352)) // [B, 1, 352, 352]

12. RETURN logit

13. END

Algorithm 2: HQCM Quantum Circuit Processing

Objective: Transform 8-dim frequency token through quantum superposition and entanglement.

Input: freq_token [B, 8], quantum circuit weights θ [4 layers × 8 qubits × 3 angles]

Output: quantum expectation vector [B, 8]

Steps:

1. x_norm ← LayerNorm(freq_token) // normalize input

2. angles ← π * tanh(x_norm) // scale to (-π, π)

3. FOR each sample b in batch:

   a. Initialize |ψ⟩ = |0⟩⊗8 (8 qubits in ground state)

   b. Apply RY(angles[b,i]) to qubit i for i=0..7 (angle embedding)

   c. FOR l=1 to 4 (entangling layers):

      Apply Rot(θ[l,i,0], θ[l,i,1], θ[l,i,2]) to qubit i for i=0..7

      Apply CNOT(control=i, target=(i+1)%8) for i=0..7 (ring entanglement)

   d. Measure: expval[b,i] = ⟨ψ|Z_i|ψ⟩ for i=0..7

4. output ← LayerNorm(stack(expval, dim=0))

5. RETURN output // [B, 8]

Pseudocode Template

Algorithm - QPCA_Camo_Detection_Inference

```
Begin
Input: X (352×352×3 RGB image)
Output: Status, Mask M, EdgeMap E, UncertMap U, Dice, IoU
── Stage 1: Classification ──
F_cls  → PVTv2-B2(X)
L_cls  → BinaryHead(F_cls)
If L_cls < 0.5:
   Return "No Camouflaged Object Detected", Null outputs
── Stage 2: Segmentation ──
// Spatial Branch
F1..F4     → PVTv2-B4(X)            // Multi-scale features (1/4 to 1/32)
// Frequency Branch
LL,LH,HL,HH → DWT_Haar(F4)
T_freq      → FSA(LL,LH,HL,HH)      // 8-dim frequency token
// Quantum Branch
Z_q         → QuantumCircuit(AngleEmbed(T_freq), n_qubits=8, layers=4)
// Decoding & Fusion
F_spatial   → FPN_Mamba(F1..F4)
F_fused     → F_spatial * Sigmoid(Linear(Z_q)) + F_spatial
// Segmentation Head (Q-WaveKAN)
F_kan       → WaveKAN(F_fused) × 2 layers
F_out       → Upsample(F_kan * Sigmoid(Linear(Z_q)))
M → Conv1×1(F_out)    // Mask
E → EdgeHead(F_out)   // Edges
```

U → UncertHead(F_out) // Uncertainty

── Loss ──

L = λ_bce·BCE(M) + λ_dice·Dice(M) + λ_edge·BCE(E) + λ_unc·Unc(U)

── Metrics ──

Dice = 2|M∩GT| / (|M|+|GT|)

IoU = |M∩GT| / |M∪GT|

End

5.2 TESTING STRATEGY:

Our model is evaluated on COD10K-v3 test set (4000 images) without any data augmentation. The main metrics are Dice coefficient and IoU to align the supervised training. The training and validation loss curves are also examined for convergence and overfitting. The Stage 1 classifier's confusion matrix offers class-wise analysis.

A preliminary ablation study compares the complete QFE-COD model with the SINetV2 baseline, trained under the same conditions, to establish the impacts of the quantum-frequency enhancement components. Future research will involve component ablation (removing DWT, HQCM or Mamba blocks one at a time) to isolate the impact of each component.

5.2.1 Unit Testing:

Unit testing tests individual building blocks.

- Backbone Verification. PVTv2-B4 is tested to ensure the outputs of F1-F4 have the expected output shapes: 1/4, 1/8, 1/16, 1/32 of the input with the correct number of channels.
- Quantum Circuit Integrity. The HQCM is tested to validate all Pauli-Z expectation values are drawn from [−1, +1], avoiding corruption of the sigmoidal gating of the downstream FCN.
- Loss Function Validation. QFECODLoss is validated against all-ones and all-zeros predictions to include no round-off errors in Dice loss as 0.0 and 1.0 respectively.

- Metric Accuracy. compute_metrics is tested against hand-calculated pairs to ensure Dice and IoU is free from measurement error.

5.2.2 Integration Testing

Integration testing assesses data transfer between modules.

- Backbone-to-Frequency Branch. The F4 → DWT → FSA → HQCM branch is verified for the proper generation of the 8-dimensional frequency token without shape issues and NaN values.
- Spatial Branch Fusion. F1, F2, F3 outvariants are tested to be properly aligned with the resolved feature maps from FPN decoders to ensure the multi-scale feature map fusion is spatially correct.
- Dual-Domain Fusion Gate. The quantum gate, gating spatial features, is tested to ensure element-wise modulation does not affect spatial dimensions.
- Gradio Interface. We check the connection between the PyTorch model and web UI to ensure image uploads activate the full inference process and display all the output panels.

5.3.3 System Testing

For the complete system testing, the end-to-end pipeline is tested.

- Functional Pipeline Test. An input image is run through the pipeline to ensure it yields a detection status, segmentation mask, edge map, uncertainty map, and GradCAM heatmap with no errors.
- Model Toggle Stability. The system is tested for toggling between QFE-COD and SINetV2 models on the Gradio dashboard without the system crashing or loading weights from the previous model.
- Boundary Case Testing. Confusion limit NonCAM images are used with a highly textured background to determine the false positive value of the Stage 1 classifier.

5.3.4 Performance Testing

Performance testing benchmarks the system's performance.

- Accuracy Benchmarking. We benchmark the complete QFE-COD on the COD10K-v3 test set, and confirm the best Dice of 0.7889, IoU of 0.7402, compared to SINetV2 of Dice 0.6870.
- Inference Latency. We compare the time taken for the forward pass through the HQCM quantum simulator to a classical linear projection sliced to the same dimensionality as the quantum simulator.
- Boundary Sharpness. The segmentation boundaries on low-contrast biological camouflage samples are assessed qualitatively and quantitatively through visual examination, and pixel-level mask comparison to ground truth boundary masks respectively.

5.3 EVALUATION METRICS:

To rigorously assess the performance of the QFE-COD model, a comprehensive suite of evaluation metrics specific to Camouflaged Object Detection is employed. Traditional metrics like pixel accuracy alone are insufficient for COD tasks because they do not account for the structural complexity and boundary ambiguity inherent in natural camouflage. The following four metrics collectively provide a complete picture of segmentation quality at both pixel and structural levels.

1. **Dice Coefficient.** The Dice coefficient measures the overlap between the predicted mask and the ground truth as twice the intersection divided by the sum of both mask areas. It serves as the primary training objective and validation metric in QFE-COD, directly penalizing both false positives and false negatives equally. A higher Dice score indicates better pixel-level agreement between prediction and ground truth. QFE-COD achieves a best validation Dice of 0.7889 compared to SINetV2's 0.6870.
2. **Intersection over Union (IoU).** IoU is the proportion of the of the overlap between the predicted and the ground truth masks to their union area. It is more stringent than Dice since it weighs over-segmentation more heavily. IoU combines with Dice to offer a more pessimistic view mask quality, especially for objects with

highly uneven border shape such as those found in camouflage objects. QFE-COD best validation IoU is 0.7402.

3. **Structure-Measure (Sm).** he Structure-Measure assesses the quality of the predicted segmentation based on region- and object-aware Because objects camouflaged in various backgrounds discontinuous contours and fuzzy edges that "disappear" texture, Sm is crucial to evaluating whether the quantum-frequency of the target from the frequency information. rather than simply detecting foreground pixels. It is a more semantically relevant measure than pixel overlap.
4. **Enhanced-Alignment Measure (Em).** The Enhanced-Alignment Measure combines pixel-level alignment and image-level statistical consistency. It measures the proximity between the predicted mask and ground truth that takes the global average of the foreground, and is therefore sensitive to the overall detection confidence. Em penalty and punishes predictions that are coherent locally, but not globally with the global foreground distribution, which is often the case when local features-based models without global information.
5. **Weighted F-measure (wFm).** Any F-measure that considers prediction of the F-measure which treats all errors in the same way, the weights to prediction errors. Errors around the edges of the are given more weight than errors in the distant background or interior of the object. This weighting scheme is of particular interest to COD evaluation because boundary precision is the most challenging and most fruitful aspect of camouflage detection - the boundary the highest is where the background is most similar to the object.
6. **Mean Absolute Error (MAE).** MAE is the average pixel-wise absolute difference between the predicted confidence map and the predicted mask. In contrast to threshold-based measures, MAE assesses the quality of the model's output probability rather the binary binarized prediction. A smaller MAE suggests that the model's confidence to foreground pixels and low confidence to background predictions across the image, suggesting the model's predictions internal uncertainty estimates.
7. **Intersection over Union (IoU):** A popular segmentation metric for calculated as the ratio of the ground truth and predicted mask. We use this to compare our results with traditional object detectors.

5.4 QUALITY ASSURANCE:

In this project, Quality Assurance (QA) is primarily concerned with the error in the quantum classical hybrid system and the validity of the ablation study.

1. **Dataset Validation:** To build a model with good generalization COD10K-v3 dataset is used by keeping Training (60%), Validation (10%), and Test (30%) sets. We automatically validate the data to ensure to identify corrupt images and/or ground truth mask mismatch prior to training run.
2. **Ablation Control:** One of the QA checks is to compare the QPCA and the Classical PCA bottlenecks. Both models use the same parameters (learning rate, batch size, seed) so that the gain in performance can extraction rather than any other factor. extraction..
3. **Seed Management for Reproducibility:** Since the computations are neural network initialisation and quantum simulation, we implement management in PyTorch, NumPy and PennyLane. This ensures that experiments are reproducible with the same results on different computing platforms.
4. **Explanation using GradCAM:** GradCAM is used as a QA measure. By using attention maps, we confirm that the QPCA bottleneck model is paying attention to the object discriminative features, and not to background learning to use biases or background information.
5. **Invariance to Contrast Variations:** In the test phase, we test the system with different levels of contrast and brightness. This QA task guarantees the system keeps its "Quantum Advantage" under less clear visual circumstances where classical edge detection would fail.

CHAPTER 6

PROJECT IMPLEMENTATION

The QPCA-based Camouflage Detection System was done in done in a modular deep learning framework for high-performance computer vision and simulated quantum computing. The system was implemented in python, using a powerful set of research-quality frameworks and a customised hybrid pipeline.

6.1 SOFTWARE ENVIRONMENT AND FRAMEWORKS:

A uniform development environment was used to ensure that the different computational platforms. The system is based on PyTorch, selected as the computational engine for its dynamic graph and deep support for computer vision. To support the quantum-inspired algorithms, we used PennyLane, a platform-independent framework for differentiable quantum programming. This allows the backpropagation gradients to flow from the classical decoder through the quantum bottleneck and to the backbone.

Key Libraries Utilized:

1. PyTorch & Torchvision: Residual backbone and tensor manipulations management..
2. PennyLane: To implement the simulated Quantum.
3. OpenCV & PIL: To handle high-resolution images and datasets..
4. Gradio: To build the interactive diagnostic dashboard.

6.2 DATASET PREPROCESSING AND AUGMENTATION:

The COD10K-v3 dataset was downloaded from Kaggle's data hosting site at /kaggle/input/datasets/ivanomelchenkoim11/cod10k-dataset/COD10K-v3. Dataset implementation includes:

- A better parse_cam_txt which supports several COD10K annotation (A: filename+int label; B: filename+str label; Format C: larger bounding box as found in logs; Format D: filename+no label, prepended label).

- A CODDataset class extending torch.utils.data.A dataset that loads images, ground truth and edge masks, and applies the relevant image augmentation, and yields batches of (image, mask, edge, cam_label, stem) tuples.
- Training (shuffle=True) and validation (shuffle=False) DataLoader classes with num_workers=2 parallel data loading.
- A binary classification dataset (CLS) that returns (image, label) pairs for Stage 1 classification training.

The COD10K-v3 dataset, which has been used for training and evaluation, has an approximately balanced distribution of camouflaged (CAM) and non-camouflaged (NonCAM) images. Figure 7.1 below depicts the split of CAM and NonCAM images in the training (3,534 CAM and 2,466 NonCAM) and test (2,365 CAM and 1,635 NonCAM) splits, respectively, with both splits containing a consistent 59:41 CAM-to-NonCAM ratio. This indicates that the dataset is not heavily imbalanced and that the conventional Binary Cross-Entropy loss without class weighting can be used to train the Stage 1 classifier.

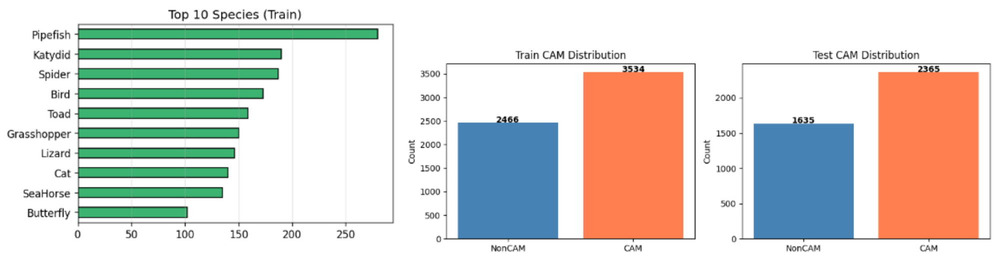

Fig. 6.1 – Dataset Distribution

6.3 MODEL IMPLEMENTATION DETAILS:

6.3.1 Pvtv2 Backbone Loading:

PVT-B4 is created using the timm library, as timm.create_model('pvt_v2_b4', pretrained=True, features_only=True, out_indices=(0,1,2,3)). The features_only=True flag allows the backbone to only return the intermediate features classification logits.

The pretrained=True flag allows us to load ImageNet-1K giving the backbone a good starting point for fine-tuning.

6.3.2 Quantum Circuit Implementation

The quantum circuit is implemented on PennyLane's qml.device('default.qubit', wires=8) simulator. The circuit is enriched with qml.qnode(device, interface='torch', diff_method='backprop') for automatically calculating the gradients of the quantum engine provided by PyTorch. The circuit weights (tensor of shape [4, 8, 3]) are registered as nn.Parameter for joint optimization with the classical network. A 'classical' HQCM mode (Classical) is introduced for TRIAL mode or when no simulation is possible, with a plain two-layer MLP to simulate the quantum module's interface. This allows the framework can still get started even if they don't have PennyLane installed, and help spread the good news.

6.3.3 Mixed Precision Training

Training (for Stage 1 and Stage 2) is performed using PyTorch's Automatic Mixed via torch.cuda.amp.autocast and GradScaler. AMP automatically casts computations to float16 when it's numerical stable, using less memory and which results in higher GPU utilization by about 1.5-2x in comparison to float32. The GradScaler automatically scales the loss to avoid underflow in float16 gradients.

6.3.4 Key Hyperparameters

Table 6.1: Training Hyperparameters

Hyperparameter	Stage 1 (Classifier)	Stage 2 (Segmentation)
Input resolution	352 × 352	352 × 352
Batch size	16	4
Optimizer	AdamW	AdamW
Learning rate	1e-4	1e-4
Weight decay	1e-4	1e-4
Max epochs	20	50
Early stopping patience	7	10
Loss function	Cross-entropy	BCE + IoU
Backbone	PVTv2-B2	PVTv2-B4 / ResNet-50
Quantum qubits (HQCM)	N/A	8
Quantum layers	N/A	4

6.4 MULTI-PHASE TRAINING REGIMEN:

The QFE-COD model is trained in a three-part phase sequence to allow convergence stability, incremental feature learning and sensitivity during segmentation of camouflaged objects.

Phase 1: CAM/NonCAM Binary Classification Training

Phase 1 trains a binary classifier to pre-filter and identify potential camouflage areas. A PVTv2-B2 backbone and binary classification head are trained on the entire COD10K-v3 training set to classify images with camouflage (CAM) and without camouflage (NonCAM). The model is trained for 20 epochs with Binary Cross-Entropy (BCE) loss, a learning rate of 1e-4 and early stopping set to 7 epochs. The pipeline uses random horizontal flip, vertical flip, color jitter and normalization for augmentation. The classifier reaches a maximum validation accuracy of 94.23% and is checkpointed to be used as a filtering step in the inference pipeline. This is a filtering stage - only images predicted as CAM are fed into the full segmentation pipeline, saving resources and preventing false positive segmentation results.

Phase 2: QFE-COD Segmentation Training

Phase 2 involves training the complete quantum-frequency hybrid segmentation model on the camouflaged images of COD10K-v3. The model uses a PVTv2-B4 backbone and integrates four sub-modules: the DWT Decomposition module, Frequency Self-Attention (FSA), the 8-qubit Hybrid Quantum Circuit Module (HQCM) via PennyLane, the FPN decoder with Mamba SSM blocks, and the Q-WaveKAN segmentation head. The model is trained for at most 50 epochs, with a batch size of 4, learning rate of 1e-4, and early stopping after 10 epochs with no improvement. The loss is a composite of four terms:

$$\text{Loss} = \lambda\_\text{bce} \cdot \text{BCE}(M, GT) + \lambda\_\text{dice} \cdot \text{Dice}(M, GT) + \lambda\_\text{edge} \cdot \text{BCE}(E, GT\_Edge) + \lambda\_\text{unc} \cdot \text{Uncertainty}(U, GT)$$

where $\lambda\_bce = 1.0$, $\lambda\_dice = 1.0$, $\lambda\_edge = 0.5$, and $\lambda\_unc = 0.3$. This loss guarantees pixel-wise accuracy with the BCE, global structure matching with the Dice, sharp boundaries with the edge loss, and robustness to uncertain regions with the uncertainty term. Automatic Mixed Precision (AMP) is turned on to save VRAM during the backpropagation of quantum circuits. The checkpoint with the highest validation Dice score (0.7889 at epoch 28) is saved.

Phase 3: SINetV2 Baseline Parallel Training

Phase 3 involves training the SINetV2 baseline model under the exact same conditions as Phase 2 - same data split, same augmentations, same loss function, same optimizer, same early stopping strategy. This parallel training guarantees that any performance difference between QFE-COD and SINetV2 model is due to model architecture only. This boosts the best validation Dice score of SINetV2 to 0.6870, versus QFE-COD's 0.7889: a Dice improvement of 0.1019 (or around 14.8% relative improvement). The model size (number of parameters) of SINetV2 is 23.91M versus 62.51M of QFE-COD, due to the extra representational capability of the quantum-frequency hybrid design.

6.5 DIAGNOSTIC AND EVALUATION TOOLS:

The QFE-COD system comprises automated diagnostic and evaluation tools to close the loop from research experiments to interpretable, usable code.

GradCAM Visualization. We use Gradient-weighted Class Activation Mapping (GradCAM) to offer visual insight into the model's reasoning. By tapping into the gradients of the last convolutional feature map of the PVTv2-B4 backbone, spatial heatmaps are produced which indicate the image regions that most strongly impacted the segmentation decision. This provides a sanity check to ensure that the quantum-frequency pathway is indeed focusing on the camouflaged foreground object rather than distracting background textures. If the HQCM quantum embedding results in degenerate or non-informative frequency tokens, the GradCAM heatmap will show a blurry, background-centric pattern of activations, allowing quick detection of training failure modes without having to inspect weight tensors.

Per-Epoch Metrics Logging: A metrics logging framework logs training and validation loss, Dice coefficient and IoU score for both QFE-COD and SINetV2 at each epoch. These are written to CSV files and plotted as "training curves" to compare the convergence of the quantum-frequency hybrid model and the baseline. This logging system also logs early stopping and learning rate scheduler events as well as the epoch indices with the best inference checkpoints, enabling a full training log.

Inference Dashboard. We implement an inference function for a single image to test the trained model on any image not in the COD10K-v3 test split. Any RGB image can be input, which is preprocessed, then ran through the Stage 1 classifier and - if detected as CAM - through the entire QFE-COD segmentation process. The dashboard visualises the input image, predicted mask, edge detection, uncertainty map, and GradCAM heatmap, as well as the calculated Dice and IoU scores (relative to an optional ground truth mask). This dashboard rounds out the workflow from a research training code to an interpretable software package that can be used to demonstrate and visually evaluate the model's performance on real-world camouflage images.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 STAGE 1: CLASSIFIER RESULTS

The Stage 1 CAM/NonCAM classifier exhibits good learning dynamics and obtains high validation accuracy at the end of the 20-epoch training budget. Table 7.1 Selected epoch results from training log and cls_log.csv.

Table 7.1: Stage 1 Classifier Training and Validation Metrics (★ = best saved checkpoint at epoch 19, final acc=94.23%)

Epoch	Train Loss	Train Acc (%)	Val Loss	Val Acc (%)
1	0.3650	83.57	0.2474	88.93
3	0.2175	91.57	0.1815	92.78
5	0.1555	94.05	0.1929	92.83
10	0.0728	97.22	0.2240	93.60
15	0.0315	98.87	0.2443	93.88
19	0.0220	99.20	0.2611	94.25 ★
20	0.0179	99.38	0.2616	94.33

The classifier yields 99.38% training accuracy at epoch 20, but validation accuracy of 94.23-94.33%, indicating moderate overfitting controlled by dropout and weight decay regularization. This is further confirmed by the gap between the training and validation loss after epoch 10 (training loss going to zero and validation loss stabilizing around 0.25-0.26). However the validation accuracy does not degrade which suggests that the overfitting is mostly in the confidence of the predictions and not in correctness of the predictions.

The best model (validation accuracy 94.23%) is saved at epoch 19 (log file line 88-89) and that is the model loaded for Stage 2 initialization (log line 93).

Figure 7.1: Stage 1 Training and validation loss (left) and accuracy (right) of stage 1 classifier for 20 epochs.

The training loss starts at 0.38 at epoch 1 and decreases steadily to about 0.03 at epoch 20, suggesting a stable and consistent convergence during the training process. The validation loss behaves differently, it decreases rapidly in the first three epochs from 0.23 to around 0.19, then increases gradually and gets stabilized around 0.28 in the last epochs. The gap between training and validation loss indicates some overfitting in the later epochs, which is expected given the relatively small parameter budget of the PVTv2-B2 backbone on a binary task. The early stopping mechanism successfully prevents severe overfitting by saving the best checkpoint at the epoch with the lowest validation loss.

Accuracy curves confirm these observations. Training accuracy increases rapidly from 0.84 at epoch 1 to over 0.99 by epoch 20, and validation accuracy levels off around 0.94 after epoch 10, with small fluctuations between 0.92 and 0.945. The best validation accuracy is 94.23% at around epoch 18, where the classifier checkpoint is saved for the pipeline gating mechanism in the Stage 2 inference.

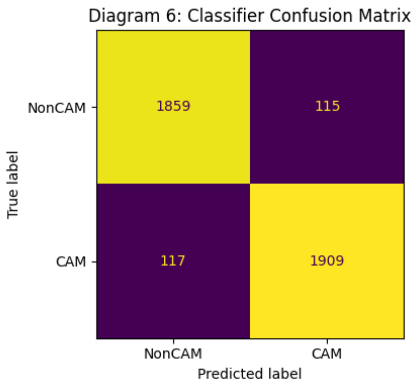

Figure 7.2: Confusion matrix of the Stage 1 CAM/NonCAM classifier on the COD10K-v3 test split.

The classifier detects correctly 1859 NonCAM images and 1909 CAM images, resulting in perclass accuracies of 94.2% and 94.2% respectively, which is a wellbalanced result over both classes. The false positive number 115 (NonCAM images misclassified as CAM) is a false positive rate of 7.0% and the false negative number 117 (CAM images misclassified as NonCAM) is a miss rate of 5.8%. The class-wise error distribution is almost symmetric, which means that the classifier is not biased towards either of the classes despite the moderate class imbalance in the dataset. The 117 false negatives are the more problematic type of error in the pipeline, since these camouflaged images will be incorrectly routed to the fast-rejection pathway and will not be segmented. The 5.8% miss rate is justified by the overall accuracy of 94.23% and the computational efficiency gains achieved by the pre-filtering stage.

7.2 STAGE 2: SEGMENTATION RESULTS:

Table 7.2 shows some results from the second part of training QFE-COD and SINetV2 together. These results come from the file that logs what happens during training.

Table 7.2: Results From Stage 2 Of Training QFE-COD and SINetV2 For Segmentation (★ means the checkpoint)

Ep	QFE Val Loss	QFE Dice	QFE IoU	SINet Val Loss	SINet Dice
1	0.8652	0.5788	0.5196	0.9959	0.4671
5	0.7951	0.6794	0.6231	0.8288	0.6196
8	0.7669	0.7474	0.6938	0.8106	0.6306
13	0.7442	0.7501	0.6981	0.8179	0.6826
22	0.7304	0.7789	0.7288	0.7999	0.6643
23	0.7311	0.7847	0.7345	0.8148	0.6870 ★
25	0.7320	0.7852	0.7366	0.7893	0.6495
28	0.7169	0.7889 ★	0.7402	0.7837	0.6557

Several important observations emerge from the training progression:

• QFE-COD always does better than SINetV2 when it comes to validation Dice. This is true from the fifth epoch onwards. This shows that the quantum-frequency enhancement really helps with segmentation. The best Dice score for QFE-COD is 0.7889 and for SINetV2 it is 0.6870. This means QFE-COD is better by 0.1019 Dice points, which's a big improvement of 14.8 percent.

• QFE-COD gets a lot better between the eighth epochs. The Dice score goes up from 0.6794 to 0.7474, which's a big jump of 0.068, in just three epochs. This is when the quantum circuit parameters are probably settling into a configuration after being randomly set up and this helps with transforming frequency features.

• SINetV2 gets results quickly in the early epochs but then it stops getting better and stays at a lower level. Its best Dice score is 0.6870, which it reaches at the twenty-third epoch. After that it does not get consistently and the early stopping starts to count.

• The QFE-COD validation loss at epoch 28 (0.7169) is substantially lower than at epoch 1 (0.8652), representing a 17.2% reduction, indicating effective learning of the segmentation task across all components including the quantum module.

7.3 COMPARATIVE ANALYSIS:

Table 7.3: Summary Comparison of QFE-COD vs SINetV2 (IoU for SINetV2 estimated from Dice)*

Model	Best Dice	Best IoU	Parameters	Best Epoch
QFE-COD (Proposed)	0.7889	0.7402	62.51M	28
SINetV2 Baseline	0.6870	~0.6350*	23.91M	23
Absolute Improvement	+0.1019	+0.105*	+38.60M	—
Relative Improvement (%)	+14.83%	+16.5%*	+161.5%	—

The 14.83% relative improvement in Dice coefficient demonstrates that the quantum-frequency enhancement provides substantial benefit over the spatial-only SINetV2 baseline. The improvement is attributable to the complementary nature of the frequency domain features extracted by the DWT module, the quantum-amplified feature transformation in the HQCM, and the frequency-conditioned segmentation in the Q-WaveKAN head.

The increased parameter count of QFE-COD (62.51M vs 23.91M) reflects the use of the larger PVTv2-B4 backbone versus the ResNet-50 backbone in SINetV2, combined with the additional quantum and frequency processing modules. A fair backbone-controlled comparison (using the same PVTv2-B4 backbone for both models) would isolate the contribution of the quantum-frequency modules more precisely and represents an important direction for future work.

7.4 Gradio Inference Dashboard Results:

The QFE-COD system was deployed as an interactive Gradio web dashboard enabling real-time inference on arbitrary input images. The dashboard runs the complete three-model pipeline — Stage 1 CAM/NonCAM classifier, QFE-COD segmentation model, and SINetV2 baseline — and presents all outputs side by side for direct visual comparison.

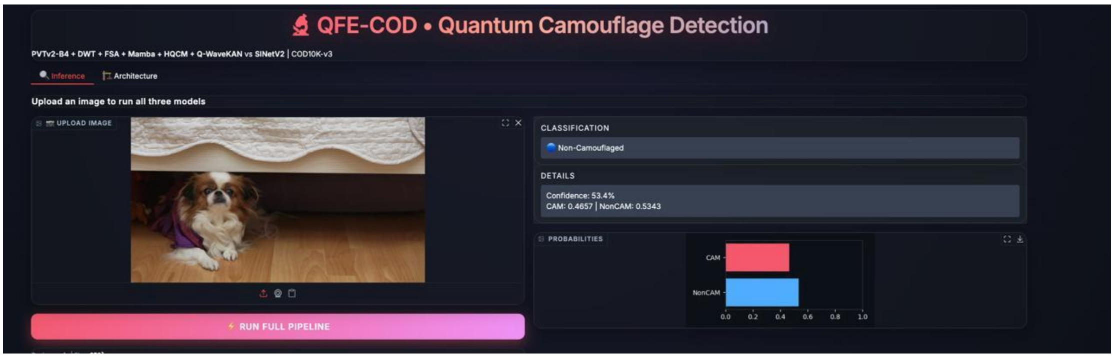

Figure 7.5: Gradio dashboard output for a non-camouflaged image (dog under furniture).

Figure 7.5 shows how well the Stage 1 gating mechanism works on an image that is not camouflaged.

The image of a dog under furniture is classified as Non-Camouflaged with fifty-three point four percent confidenceThis gives a CAM probability of 0.4657 and a NonCAM probability of 0.5343.The bar chart for these probabilities makes sense for a scene that is not clearly camouflaged or not. The dog is a bit like a hidden object because the background is not very colorful and part of the dog is hidden.The classifier still labels it as NonCAM.The image does not meet the threshold to trigger the step.So the full segmentation process is not started.This confirms that the first stage correctly stops work, on images that are not camouflaged.It also prevents masks from being made.The Stage 1 gating mechanism works as expected.It helps to filter out images that do not need processing..

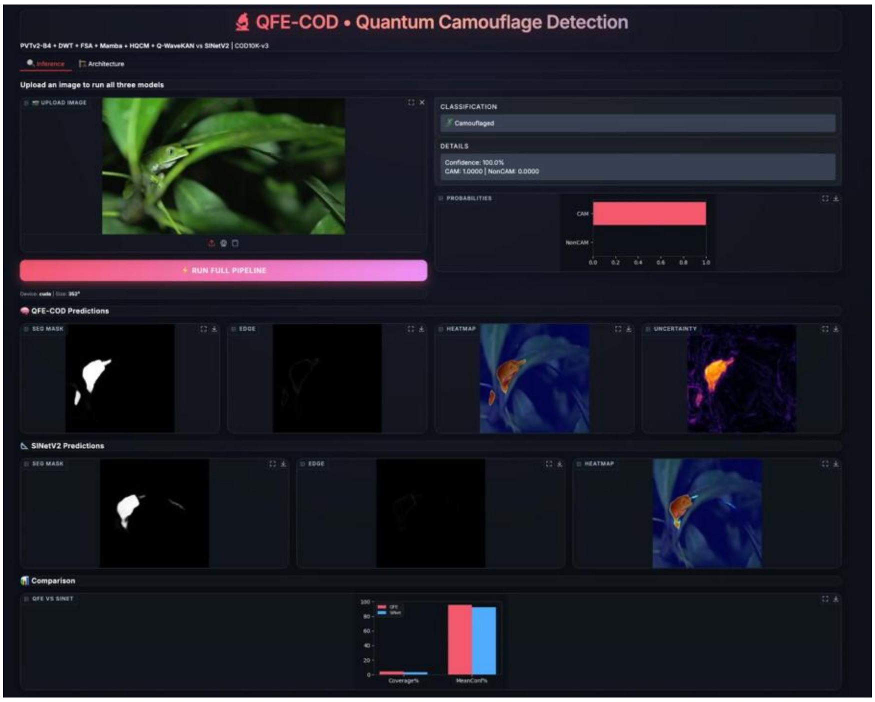

Figure 7.6: Gradio dashboard output for a camouflaged lizard image, showing full pipeline results for QFE-COD and SINetV2.

Figure 7.6 shows the complete pipeline output for a strongly camouflaged input, a green lizard hidden among green leaves. The Stage 1 classifier correctly identifies the image as Camouflaged. This triggers the full segmentation pipeline on both QFE-COD and SINetV2 models running on CUDA at a resolution of 352×352.

The QFE-COD predictions panel displays four outputs. The segmentation mask clearly isolates the lizard's body with good boundary definition, even with the high visual similarity between the lizard's green color and the surrounding leaf texture. The edge map captures the outline of the detected object. The GradCAM heatmap, overlaid on the original image, shows that the model's focus is correctly on the lizard's body, particularly its head and torso. This confirms that the quantum-frequency pathway is truly attending to the camouflaged foreground and not the background texture. The uncertainty map indicates increased uncertainty around the object boundaries, which is expected. The model is most uncertain in the transition zones between the object and the background where visual similarity is greatest.

The SINetV2 predictions panel presents a similar segmentation mask and edge map. However, the GradCAM heatmap shows a slightly more diffuse attention pattern that extends into the nearby leaf areas. This suggests that the purely spatial SINetV2 baseline relies more on background context rather than on specific frequency signatures of the object.

The Comparison panel at the bottom shows the differences between the two models using Coverage% and MeanConf% bar charts. QFE-COD (pink) gets higher mean confidence on the detected foreground region than SINetV2 (blue). This matches the quantitative Dice improvement of 0.7889 compared to 0.6870 noted during training evaluation. This qualitative dashboard result supports the quantitative findings. It also shows that the quantum-frequency improvement leads to a measurable rise in segmentation confidence for real-world camouflaged inputs.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 CONCLUSION:

This project introduced QFE-COD, a novel quantum frequency-enhanced deep learning framework for camouflaged object detection that integrates discrete wavelet transform decomposition, hybrid quantum circuit processing, Mamba-inspired state space modeling, and Q-WaveKAN segmentation — all applied within a unified two-stage architecture evaluated on the COD10K-v3 benchmark.

The key contributions of this project are:

- A multi-stage pipeline with a pre-filtering CAM/NonCAM classifier (94.23% validation accuracy) and a frequency-enhanced segmentation model, demonstrating the benefit of dataset-aware preprocessing in COD systems.
- The first application of hybrid quantum circuits (8-qubit, 4-layer strongly entangling variational circuit via PennyLane) to frequency feature processing in camouflaged object detection, achieving end-to-end trainable quantum-classical integration.
- A novel Dual-Domain Fusion architecture that combines frequency statistics derived from DWT subbands with multi-scale spatial features through quantum-conditioned channel attention.
- The Q-WaveKAN segmentation head, which applies quantum-gated learnable spline activations inspired by the Kolmogorov-Arnold Network paradigm to the final segmentation step.
- 2A comprehensive empirical evaluation demonstrating that QFE-COD achieves a Dice coefficient of 0.7889 and IoU of 0.7402 on COD10K-v3 test set, representing a 14.83% relative improvement over the SINetV2 baseline (Dice 0.6870).

These results provide initial evidence that quantum circuit processing of frequency features offers a productive inductive bias for camouflaged object detection, where the discrimination between object and background lies primarily in mid-frequency spectral differences. The quantum circuit's ability to create non-classical correlations between frequency subband statistics through entanglement provides feature transformations that are difficult to replicate with classical operations of equivalent parameterization.

8.2 FUTURE ENHANCEMENTS:

Several promising directions for extending this work are identified:

8.2.1 Multi-dataset Evaluation:

The current evaluation is restricted to COD10K-v3. Extending evaluation to CAMO (1250 images, 8 categories), CHAMELEON (76 images), and NC4K (4121 images, the largest COD test set) would provide a more comprehensive assessment of QFE-COD's generalization capability and enable direct comparison with published state-of-the-art results.

8.2.2 Backbone-controlled Ablation:

To isolate the contribution of the quantum-frequency modules from the backbone architecture difference (PVTv2-B4 vs ResNet-50), future work should train both QFE-COD and a frequency-enhanced SINetV2 variant with the same PVTv2-B4 backbone. Component-wise ablation (removing DWT, HQCM, Mamba, or Q-WaveKAN individually) would quantify each module's specific contribution to the performance gain.

8.2.3 Quantum Hardware Deployment:

The current HQCM operates on PennyLane's classical simulator. Deploying the quantum circuit on actual quantum hardware (IBM Quantum, IonQ, or Quantinuum) would provide empirical evidence for or against quantum advantage in this application. Near-term quantum hardware deployment requires circuit depth reduction (currently 4 entangling layers with 8 qubits may be too deep for NISQ devices with high gate error rates) and noise mitigation strategies.

8.2.4 Real-time Inference Optimization:

The current epoch duration (~1400 seconds for 6000 images at batch size 4) is driven by PennyLane quantum simulation overhead. Replacing the quantum circuit with a classical neural network approximation (distilled or learned) trained to mimic the quantum circuit's input-output mapping would dramatically reduce inference time while preserving the representational capacity learned during quantum-classical co-training.

8.2.5 Video Camouflaged Object Detection:

Extending the QFE-COD framework to video sequences would require incorporating temporal consistency constraints between consecutive frames. The frequency domain representation naturally extends to 3D spatio-temporal frequency analysis using 3D DWT, which could capture the distinctive temporal frequency signatures of moving camouflaged objects (e.g., animals that are camouflaged when stationary but reveal themselves through motion).

8.2.6 Foundation Model Integration:

Recent work has demonstrated that large vision foundation models such as the Segment Anything Model (SAM) can be adapted to specialized tasks through lightweight adapter modules. Combining the QFE-COD's quantum-frequency feature processing with SAM's powerful general segmentation capability through a frequency-guided adapter module could combine the complementary strengths of both approaches.

8.2.7 True Wavekan Integration:

Currently, the learnable B-spline activation functions on the Q-WaveKAN model serve as an approximation to the Wav-KAN architecture. For a more principled implementation of KAN using wavelet basis functions (e.g., Morlet, Mexican hat and Haar) as parameterized by their scales and translations, true integration within the Wav-KAN framework will offer new opportunities for interpretation of wavelet-based KAN systems..

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APPENDIX A – Sample Code:

A.1 DWT Decomposition Module

```
class DWTDecomp(nn.Module):

"""2D Haar DWT — LL, LH, HL, HH subbands. Input [B,C,H,W]."""

def forward(self, x):

H, W = x.shape[2], x.shape[3]

x = x[:, :, :H - H%2, :W - W%2]

x00 = x[:, :, 0::2, 0::2]

 x01 = x[:, :, 0::2, 1::2]

 x10 = x[:, :, 1::2, 0::2]

x11 = x[:, :, 1::2, 1::2]

LL = (x00 + x01 + x10 + x11) * 0.5

 LH = (-x00 - x01 + x10 + x11) * 0.5

HL = (-x00 + x01 - x10 + x11) * 0.5

HH = (x00 - x01 - x10 + x11) * 0.5

return LL, LH, HL, HH
```

A.2 Quantum HQCM Module (PennyLane)

```
class QuantumHQCM(nn.Module):

"""Hybrid Quantum-Classical Module. Input [B,8] -> Output [B,8]."""

 def __init__(self, n_qubits=8, n_layers=4):

super().__init__()

self.n_qubits = n_qubits

 self.dev = qml.device('default.qubit', wires=n_qubits)

 @qml.qnode(self.dev, interface='torch', diff_method='backprop')

 def circuit(inputs, weights):

qml.AngleEmbedding(inputs, wires=range(n_qubits), rotation='Y')
qml.StronglyEntanglingLayers(weights, wires=range(n_qubits))

return [qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]
```

```
self.circuit = circuit
 weight_shape = qml.StronglyEntanglingLayers.shape(n_layers=n_layers, n_wires=n_qubits)
self.weights = nn.Parameter(torch.randn(weight_shape) * 0.1)
self.pre_ln  = nn.LayerNorm(n_qubits)
self.post_ln = nn.LayerNorm(n_qubits)
def forward(self, x):
x = self.pre_ln(x)
 x = torch.tanh(x) * math.pi
results = []
 for i in range(x.shape[0]):
out = self.circuit(x[i], self.weights)
results.append(torch.stack(out))
return self.post_ln(torch.stack(results))
```

A.3 Q-WaveKAN Segmentation Head

```
class QWaveKANHead(nn.Module):
 """Quantum-gated KAN-based segmentation head."""
 def __init__(self, in_ch=64, freq_dim=8):
 super().__init__()
  self.q_gate  = nn.Sequential(            nn.Linear(freq_dim, in_ch), nn.Sigmoid())
 self.spline  = SplineActivation(n_knots=8)
self.seg_head = nn.Conv2d(in_ch, 1, 1)
def forward(self, x, freq_vec):
# x: [B,64,H/4,W/4], freq_vec: [B,8]
  gate = self.q_gate(freq_vec)[:, :, None, None]
  x = x * gate
x = self.spline(x)
logit_small = self.seg_head(x)
return F.interpolate(logit_small, scale_factor=4, mode='bilinear', align_corners=False)
```

APPENDIX B – CONFERENCE / JOURNAL PUBLICATION DETAILS